The observed temperature power spectrum through fourth order in a radiation-only universe: final report

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Abstract

We compute the observed temperature anisotropy through third order, and the angular power spectrum through fourth order, in the primordial amplitude ϵ. The setting is a flat radiation-only universe in general relativity, with Planck test photons emitted on $T_{\rm phys} = T_\star$ and an observer on a fluid worldline.

The calculation uses the fluid's velocity potential and comoving labels as coordinates (*temperature frame*). In these coordinates the observed temperature is an exact line integral of the shear of a conformally rescaled metric, and the observer's own temperature cancels exactly in the normalised map. Further results:

- The spatial Einstein equations take a compact mixed trace-free form, which yields a triangular solution scheme with elementary Green functions at every order.
- This scheme satisfies all ten field equations at every order (proved).
- All second-order field kernels are given in closed form for general momenta, in the entire functions Si, Cin and $\mathfrak{e}_m$. This makes every resonance surface manifestly regular.
- Third-order kernels are explicit nested integrals.
- The complete map $\Theta^{(1,2,3)}$ is constructed. It retains photon propagation, emission surface, observer clock, tetrad and normalisation terms.
- The linear transfer function T_ℓ is given in closed form for every ℓ, as a short sum of the same entire functions J_m used at second order. The coefficients are Laurent coefficients of two explicit rational functions, with no logarithms or Ci.
- $C_\ell^{(22)}$ and $C_\ell^{(13)}$ are reduced to exact finite-dimensional integrals, with every sky integral done in closed form. On the monochromatic shell the answer is organised as $C_\ell = 4\pi v T_\ell^2 + v^2[Q_\ell + 4\pi T_\ell \,\mathrm{Re}\, R_\ell]$, with $v = \epsilon^2\sigma^2$. Two exact symmetry reductions simplify it. The second-order metric reduces to two amplitudes, one scalar and one tensor polarisation, with no vector part. The angle-averaged cubic metric response is a single scalar.
- The fourth-order spectrum splits exactly into three parts. They are defined by the centred log-temperature map L, which is the log of $T_{\rm obs}$ minus its sky mean. The three parts are the fourth-order spectrum of L, a lognormal normalisation term given by an explicit $3j$ series in $C^{(11)}$ alone, and a collapsed bispectrum of L. On the monochromatic shell, all first-order quantities are local in the primordial field. The resonant integrals over angle and Green-function times are confined to the second- and third-order metric response.
- The finite-time map admits a compact centred-exponent form, and the spectrum is the sum of a corrected linear-response norm and a quadratic-response norm. These are exact reorganisations through the required order, not evaluations of the remaining integrals.
- In the superhorizon limit the observable is a local trace-free-curvature quadrupole, *exact in* ζ. Its leading-gradient Gaussian spectrum factorises into an observer dilation and a curvature norm:

$$C_{2,\rm lead} = \tfrac{4\pi}{225}\epsilon^2\sigma^2\,\langle\Xi^2_{\rm phys}\rangle\big(1+\tfrac{5}{3}\epsilon^2\sigma^2\big), \qquad \langle\Xi^2_{\rm phys}\rangle = \langle e^{-4\zeta_o}\rangle\Xi^2 = e^{8\epsilon^2\sigma^2}\Xi^2, \qquad \Xi \equiv x_o^2 - x_\star^2.$$

Through fourth order this gives the factor $1+\frac{29}{3}\epsilon^2\sigma^2$, with $\frac{29}{3} = 8+\frac{5}{3}$. The 8 is the Gaussian average of the observer's own random local dilation; it is part of the answer for the specified ensemble. The $\frac{5}{3}$ is the gradient-squared curvature. For any nonnegative spectrum, the leading-gradient coefficients lie in $[8, \frac{29}{3}]\, m_0$ for the quadrupole and $[18, 46]\, m_0$ for the octupole. The monochromatic shell attains both upper bounds.

- The leading octupole is a local curvature-gradient octupole, $\Theta_{3,\mathrm{lead}} = \beta_3 \nabla_{\langle a} R_{bc\rangle} n^a n^b n^c$. Its spectrum also resums. Through fourth order it gives $C_3 = \epsilon^2 C_3^{(11)}(1+46\epsilon^2\sigma^2)$ at $O(k_0^6)$.
- These statements concern the leading-gradient expressions, not the finite-wavelength solution.

Analytical proofs and exact computer-algebra checks (no numerical evidence) are distinguished in the verification ledger. We state explicitly which integrals remain unevaluated and which limits remain open.

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1 Problem and summary of results

The task (`problem_statement.tex`) asks for $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}] + O(\epsilon^6)$ for the observed map

$$\Theta(\hat{n}) = \frac{T_{\rm obs}(\hat{n})}{\frac{1}{4\pi}\int \mathrm{d}\Omega'\, T_{\rm obs}(\hat{n}')} - 1, \qquad T_{\rm obs} = T_\star \frac{E_o}{E_e}, \quad E = -u_\mu p^\mu.$$

The physical model is a perfect radiation fluid with $P = \rho/3$, $\rho = \alpha T_{\rm phys}^4$ and $\Lambda = 0$, with Gaussian adiabatic primordial perturbations and no independent higher-order modes. The calculation must be analytical and keep every term, with all normalisations fixed. A monochromatic specialisation is to be made only at the end.

Requirement	Result (section; status)
Gaussian variable; no higher-order modes	$\zeta = \frac{1}{6}\ln\det\gamma_{ij}$, exactly conserved; Def. 4 (§4)
Constrained initial data at every order	regular solution at any t_i; constraints proved (Thm. 5); forced data nonzero
Field solution through ϵ^3 incl. vectors/tensors	triangular scheme (§5); 2nd order [closed form]; 3rd order [explicit nested integrals]
Observer, clock, tetrad, emission surface	§6; explicit Fermi–Walker rotation, irrelevant to the spectrum at this order (Prop. 8)
Complete $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$	§6, with an independent generator and Ward checks
$C_\ell^{(11)}$	[closed form, every ℓ]: entire-function T_ℓ (Prop. 11), proved for $\ell = 2, \ldots, 6$
$C_\ell^{(22)}$, $C_\ell^{(13)}$	[exact finite-dimensional integrals], organised as $C_\ell = 4\pi v T_\ell^2 + v^2[Q_\ell + 4\pi T_\ell\,\mathrm{Re}\,R_\ell]$ ((44)); sky integrals evaluated (§8); shell metric reduced to two amplitudes, contracted cubic response scalar; existence proved for finite times and compactly supported spectra; log-map decomposition with the lognormal part [explicit] (convergent $3j$ series, or one Legendre integral of ξ^2; Thm. 12)
Monochromatic, held-fixed quantity	$\langle g^2\rangle = \sigma^2$ fixed (§8.4)
Asymptotic limits	superhorizon: [closed form], including an all-amplitude Gaussian average of the leading-gradient map, factorised into the random observer dilation $\times$ a curvature norm ((49)); leading octupole [closed form] (§9.2); late time: [first order]; general k_0 at fourth order: [unevaluated integrals] (§9)
Convergence for general spectra	compact support: [proved]; IR: leading cancellation [derived] from the Ward identities; UV: [power counting only]

Requirement	Result (section; status)
Verification	§10 (exact symbolic, generic-jet proofs)

2 Conventions and the experiment

Signature $(-,+,+,+)$, $c=1$. The background is $a=a_1 t$ and $\bar{T}=\mathcal{T}/(a_1 t)$, with $8\pi G\alpha\mathcal{T}^4=3a_1^2$.

Fourier and perturbative conventions.

- $g(X)=\int_k g_k e^{ik\cdot X}$, with $\int_k=\int \mathrm{d}^3k/(2\pi)^3$ and $\langle g_k g_{k'}\rangle=(2\pi)^3\delta^3(k+k')P(k)$; also $\mathcal{P}(k)=k^3P/2\pi^2$.
- Every field is $Y=\sum_r \epsilon^r Y^{(r)}/r!$ with $Y^{(r)}=\int_{k_1\ldots k_r} K_r(k_1,\ldots,k_r)\, g_{k_1}\cdots g_{k_r} e^{iK\cdot X}$ and $K=\sum k_a$. The symmetric kernels carry no extra $1/r!$.
- *Multilinear rule:* K_r is the coefficient of $\epsilon^r z_1\cdots z_r$ in the *full* series Y when $g=\sum_a z_a e^{ik_a\cdot X}$ is inserted and only monomials in distinct labels are kept. Equivalently, the coefficient of $z_1\cdots z_r$ in $Y^{(r)}$ is $r!\,K_r$, as the integral representation above requires. This fixes every combinatorial factor; for example $K_2(k,0)=-k\cdot\partial_k K_1$ in §7.
- Spherical harmonics $Y_{\ell m}$ are complex, with the Condon–Shortley phase.

Measurement labels.

- *Emission:* the surface $T_{\rm phys}=T_\star$. Emission is isotropic and Planckian in the fluid rest frame. The intersection is the branch continuously connected to the background one.
- *Observer:* the fluid line $X=0$. The initial event is $T_{\rm phys}=T_I>T_\star$ on that line, and observation follows after fixed proper time τ_o.
- *Tetrad:* Fermi–Walker transported, with initial orientation $\gamma(t_I,0)^{-1/2}$ (the symmetric square root, aligned with the marked labels).
- *Background times:* $\bar{t}_\star=\mathcal{T}/(a_1T_\star)$, $\bar{t}_I=\mathcal{T}/(a_1T_I)$ and $\bar{t}_o^2=\bar{t}_I^2+2\tau_o/a_1$. We write $x\equiv kt/\sqrt{3}=k/(\sqrt{3}\,aH)$.

3 Exact framework

Lemma 1 (Temperature frame)**.** *For an irrotational radiation fluid, $Tu_\mu=-\mathcal{T}\partial_\mu t$ defines the velocity potential t. In coordinates (t,X^i), with comoving labels X^i,*

$$g=N^2\big(-\mathrm{d}t^2+\gamma_{ij}\mathrm{d}X^i\mathrm{d}X^j\big),\qquad N=\mathcal{T}/T_{\rm phys},\qquad u=N^{-1}\partial_t$$

exactly. Euler's equation holds identically, and continuity is equivalent to $\partial_t\det\gamma=0$. Hence $\gamma=e^{2\zeta(X)}\exp H$ with $\operatorname{tr}H=0$, and the shear $\sigma_{ij}=\frac{1}{2}\partial_t\gamma_{ij}$ is traceless.

Lemma 2 (Exact redshift)**.** *Null geodesics of g and of $\tilde{g} = g/N^2$ coincide, and $E = \tilde{E}/N$. Hence*

$$T_{\rm obs}(\hat{n}) = T_{\rm fluid}(o)\, e^{-I(\hat{n})}, \qquad I = \int_{t_e(\hat{n})}^{t_o} \sigma_{ij}(t, X(t))\, e^i e^j \, \mathrm{d}t,$$

where $\mathrm{d}X^i/\mathrm{d}t = e^i$, $\gamma_{ij} e^i e^j = 1$ and $\mathrm{d}e^i/\mathrm{d}t = -2S^i{}_j e^j - \Gamma^i{}_{jk} e^j e^k + e^i \sigma_{jk} e^j e^k$, with $S = \gamma^{-1}\sigma$. The emission event is the root of $\ln t_e + \nu(t_e, X(t_e)) = \ln \bar{t}_\star$, with $N = a_1 t e^\nu$.

$T_{\rm fluid}(o)$ is direction-independent. It therefore cancels from Θ *exactly*. The emission-surface temperature shift, Sachs–Wolfe, Doppler and integrated Sachs–Wolfe terms are all contained in the single integral I.

Proposition 3 (Unified field equations)**.** *With $n = \ln N$, Einstein's equations are equivalent to*

$$(N) \quad 3\nu'' - \Delta_\gamma \nu - 2|D\nu|^2 + \operatorname{tr} S^2 + \tfrac{3}{t^2}(e^{-2\nu} - 1) = 0, \tag{1}$$

$$(T) \quad \mathrm{TF}\left[S' + 2(t^{-1} + \nu')S + R^k{}_j[\gamma] - 2D^k D_j \nu + 2D^k \nu D_j \nu\right] = 0, \tag{2}$$

$$(M) \quad D_j S^j{}_i - 2\partial_i \nu' + \tfrac{2}{t}\partial_i \nu + 2S^j{}_i \partial_j \nu + 2\nu' \partial_i \nu = 0, \tag{3}$$

$$(\mathrm{Ham}) \quad R - 4\Delta\nu - 2|D\nu|^2 - \operatorname{tr} S^2 + 6(t^{-1} + \nu')^2 - 6e^{-2\nu}/t^2 = 0. \tag{4}$$

In the mixed form (T), the $\sigma\sigma$ terms of the spatial equation cancel, because $\gamma^{ki}\sigma'_{ij} = (S^k{}_j)' + 2S^k{}_i S^i{}_j$. This is proved for generic γ (six free functions, t-independent determinant) and arbitrary N, on independent symbolic 2-jets.

4 Primordial state and initial data

Definition 4 (Gaussian variable; absence of independent higher-order modes)**.** The state is the solution of Prop. 3 with the following properties:

(G) $\zeta = \epsilon g$ exactly, with g a zero-mean, homogeneous, isotropic Gaussian field;

(T) $\gamma \to e^{2\zeta}\delta_{ij}$ as $t \to 0$. The limit is conformally flat in the marked conformally Cartesian material chart, so there is no tensor seed;

(V) $\mathrm{d}(Tu) = 0$, so there is no vorticity seed;

(R) ν and H are even and analytic in t at fixed momenta, with $\nu(0) = 0$ and $H(0) = 0$, so there are no decaying modes.

"No independent higher-order modes" means $\zeta^{(r\geq 2)} \equiv 0$ together with (T), (V) and (R) at every order.

Because $\det\gamma$ is t-independent, $\zeta = \frac{1}{6}\ln\det\gamma$ at *every* time. As $t \to 0$, constant-t slices become uniform-density slices, so ζ is the nonlinear uniform-density (δN) curvature perturbation.

The residual conformal relabellings are fixed as follows: the dilation by $\langle g\rangle = 0$; special conformal transformations by statistical homogeneity; translations and rotations are statistical symmetries.

Exactly in ζ, the gradient-expansion data are

$$\nu = \tfrac{t^2}{18} e^{-2\zeta}\left(2\nabla^2\zeta + |\nabla\zeta|^2\right) + O(t^4), \quad H_{ij} = \tfrac{t^2}{3} e^{-2\zeta}\ \mathrm{TF}\left[\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta\right] + O(t^4). \tag{5}$$

The initial data at any t_i are the regular solution of §5 evaluated at t_i, with $\rho = 3a_1^2/(8\pi G N^4)$ and $u = N^{-1}\partial_t$. Higher-order data are nonzero. For example, $\gamma^{(2)}(0) = 4g^2\delta$, and the $\nu^{(2)}$ kernel is $\frac{t^2}{9}[2(k_1^2 + k_2^2) - k_1 \cdot k_2] + O(t^4)$. The forced $K = 0$ mode ($5k^2t^2/9$ at $k_2 = -k_1$) is kept. The data evolve back into the same regular solution, so t_i drops out identically.

5 Field solution at every order

5.1 Triangular scheme

At total momentum K, the order-r kernels (ν_r, A_r), with A_r the order-r part of H, solve $\mathcal{L}_K[\nu_r, A_r] + c^{(r)} = 0$. The operators are the same at every order:

$$L_N = 3\nu'' + K^2\nu - \tfrac{6}{t^2}\nu, \qquad (L_M)_i = \tfrac{i}{2}K_jA'_{ji} - 2iK_i(\nu' - \nu/t),$$
$$L_T = \tfrac{1}{2}(A'' + \tfrac{2}{t}A') + \mathrm{TF}\, R_1[A] + 2\,\mathrm{TF}(K_iK_j)\nu.$$

Here $R_1[A]_{ij} = \frac{1}{2}(K^2A_{ij} - K_aK_iA_{aj} - K_aK_jA_{ai})$. The source $c^{(r)}$ is the multilinear coefficient with $(\nu_r, A_r) = 0$; it is explicit in the lower orders.

Write $c = K/\sqrt{3}$, $x = ct$, $u_1 = \sin x/x - \cos x$ and $u_2 = -\cos x/x - \sin x$. The coefficient c_r is the t^2 coefficient of (5) at order r. In particular $c_2 = [2(k_1^2 + k_2^2) - k_1 \cdot k_2]/9$ and $c_3 = \frac{2}{9}[-2\sum k_a^2 + \sum_{a<b} k_a \cdot k_b]$. Then

$$\nu_r = \frac{9c_r}{K^2}u_1(x) - \frac{1}{3c}\Big[u_2(x)\int_0^t u_1(cs)c_N^{(r)}\mathrm{d}s - u_1(x)\int_0^t u_2(cs)c_N^{(r)}\mathrm{d}s\Big], \tag{6}$$

$$h_S = \tfrac{3}{2K^2}\big[4K^2\nu_r + 12\nu_r'/t + 12\nu_r/t^2 + c^{(r)}_{\mathrm{Ham}}\big] \quad \text{(algebraic, from (Ham))}, \tag{7}$$

$$w = \tfrac{2i}{K^2}\int_0^t P_\perp c_M^{(r)}\mathrm{d}s, \qquad h = -2\int_0^t \tfrac{s}{t}\tfrac{\sin K(t-s)}{K}\,\mathrm{TT}[c_T^{(r)}](s)\,\mathrm{d}s, \tag{8}$$

$$A_r = (\hat{K}\hat{K} - \tfrac{1}{3}\delta)h_S + (K \otimes w + w \otimes K) + h. \tag{9}$$

At $K = 0$ no projector appears: $A_r = -2\int_0^t \frac{s}{t}(t-s)c_T^{(r)}(s)\mathrm{d}s$ and $\nu_r = c_r t^2 - \frac{1}{3}\int_0^t \frac{t^3-s^3}{3ts}c_N^{(r)}\mathrm{d}s$. The lapse particular part is $O(t^4)$; the spatial solution generally starts at $O(t^2)$. The free regular lapse coefficient c_r is fixed by the leading, $O(t^0)$ Hamiltonian constraint, equivalently by the $O(t^2)$ lapse data in (5).

Theorem 5 (Consistency at every order)**.** *The triangular solution satisfies all ten field equations at every order.*

Proof. Let $\mathbb{E}_{\mu\nu} = G_{\mu\nu} - 8\pi G T_{\mu\nu}$. Matter conservation is exact (Lemma 1), so $\nabla_\mu \mathbb{E}^\mu{}_\nu = 0$. Inductively assume that all residuals vanish below order r. At order r this is the background Bianchi identity, with $\mathcal{H} = 1/t$. Set $f = \mathbb{E}^t{}_t$, $v_i = \mathbb{E}^t{}_i$, and $B^j{}_i = \mathbb{E}^j{}_i$, suppressing the order label. Equation (N), which is solved by construction, gives $f = \frac{1}{2}\mathbb{E}$, hence $\mathrm{tr}\, B = f$. The two Bianchi identities are therefore (at this order B is symmetric, since all lower residuals vanish)

$$f' + \frac{2}{t}f - iK_iv_i = 0, \qquad v_i' + \frac{4}{t}v_i + iK_jB^j{}_i = 0. \tag{10}$$

For $K \neq 0$, the scalar spatial solution imposes (Ham), so $f = 0$; the vector solution imposes $P_\perp v = 0$. The first identity in (10) now gives $K \cdot v = 0$, and thus the *longitudinal* momentum constraint follows rather than being assumed. Consequently $v = 0$, $\mathrm{tr}\, B = 0$, and the second identity gives $K_jB^j{}_i = 0$. The tensor solution imposes $\mathrm{TT}[B] = 0$. A symmetric tensor that is trace-free and transverse is itself TT, so $B = 0$.

For $K = 0$, the full trace-free spatial equation is solved, and (10) gives $v_i = b_i/t^4$ and $f = b/t^2$. The regular field series has $(M)_i = O(t)$, hence $v_i = O(t^{-1})$; this excludes $b_i \neq 0$. It does *not* exclude $b \neq 0$. Here the initial Hamiltonian constraint is essential: the coefficient c_r from (5) cancels the $O(t^0)$ term of (Ham). Since $\mathbb{E}^t{}_t = -(\mathrm{Ham})/(2N^2)$ and $N \sim a_1 t$, this gives $\lim_{t\to 0} t^2 f = 0$, hence $b = 0$. Finally $\mathrm{tr}\, B = f = 0$ and $\mathrm{TF}\, B = 0$ give $B = 0$. This completes the induction. □

Remark 6 (Why initial constraint matching cannot be replaced by regularity). At linear order and $K = 0$, take $\nu = \beta t^2$, $H = 0$, $\zeta = 0$ with arbitrary constant β. Then (N), (M) and (T) hold, but (Ham) $= 36\beta$. The metric variables are even and analytic. This mode is excluded by (5), not by analyticity alone.

5.2 Compact field-source representation

The large component expressions in the ancillary files need not be the working form of the answer. Keep the matrix products and the scalar/vector/tensor projections factored. In particular, put $M = e^H$, $\chi = \zeta + 2\nu$, and let ∇_M and $\mathrm{Ric}[M]$ refer to the spatial metric M. The conformal transformation of spatial Ricci curvature and of the lapse Hessian gives the exact identity

$$\begin{aligned}\mathcal{F} &\equiv \mathrm{TF}\left[R^i{}_j[\gamma] - 2D^iD_j\nu + 2D^i\nu D_j\nu\right] \\ &= e^{-2\zeta}\,\mathrm{TF}\big\{M^{-1}\left[\mathrm{Ric}[M] - \nabla_M \mathrm{d}\chi + \mathrm{d}\chi\otimes\mathrm{d}\chi - 2\mathrm{d}\nu\otimes\mathrm{d}\nu\right]\big\}\,.\end{aligned} \tag{11}$$

Here $\nabla_M \mathrm{d}\chi$ is the covariant Hessian. Thus (T) is simply $S' + 2(t^{-1} + \nu')S + \mathcal{F} = 0$. Its matrix shear can be generated through cubic order as

$$S = \frac{1}{2}H' + \frac{1}{4}[H', H] + \frac{1}{12}[[H', H], H] + O(\epsilon^4). \tag{12}$$

These commutators must not be dropped: matrices at different orders or momenta need not commute. Together with (N), (M), (Ham) and the Green operators above, these identities give a compact source prescription at arbitrary momenta: retain lower-order fields, set the unknown top-order kernels to zero, and extract the multilinear coefficient. No momenta, endpoint terms or tensor sectors are discarded. The ancillary files contain the expanded coefficients of this factored representation. This is a structural simplification, not a claim that the remaining finite-time integrals collapse to a smaller set of evaluated scalar functions.

5.3 Explicit kernels

First order [closed form].

$$\nu_1 = -(\sin x/x - \cos x), \qquad H_1 = -\tfrac{6}{k^2}\big(\tfrac{2\sin x}{x} - \cos x - 1\big)\big(k_ik_j - \tfrac{1}{3}\delta_{ij}k^2\big), \qquad \zeta_1 = 1.$$

Second order [closed form, general momenta]. Every integral above is of the form $\int_0^t B(s)\mathrm{d}s$, where $B = \sum_\omega e^{i\omega s}R_\omega(s)$ and each R_ω is a Laurent polynomial. The frequencies ω are integer combinations of $p/\sqrt{3}$, $q/\sqrt{3}$, $K/\sqrt{3}$ and K (with $p = |k_1|$, $q = |k_2|$, $K = |k_1 + k_2|$); they vanish on resonance surfaces. We integrate term by term in *entire* functions:

$$\int_0^t s^m e^{i\omega s}\mathrm{d}s = t^{m+1}\mathfrak{e}_m(i\omega t), \qquad \mathfrak{e}_m(z) = \int_0^1 u^m e^{zu}\mathrm{d}u \qquad (m \geq 0), \tag{13}$$

$$\int_0^t s^{-M}\Big[e^{i\omega s} - \sum_{j<M}\tfrac{(i\omega s)^j}{j!}\Big]\mathrm{d}s = J_M(\omega, t), \quad J_1 = -\,\mathrm{Cin}(\omega t) + i\,\mathrm{Si}(\omega t), \tag{14}$$

$$J_M = \tfrac{t^{1-M}E_M}{1-M} + \tfrac{i\omega}{M-1}J_{M-1}, \qquad E_M = e^{i\omega t} - \sum_{j<M}\tfrac{(i\omega t)^j}{j!}. \tag{15}$$

Here $\mathrm{Cin}(x) = \int_0^x (1 - \cos u)u^{-1}\mathrm{d}u$ is entire and even, and Si is entire and odd. The subtracted Taylor pieces sum to the principal part of B at $s = 0$, which vanishes (this is asserted in the code). *No division by any frequency occurs*, so the kernels are manifestly finite on every resonance surface:

- the scalar collinear surfaces $\mu = \pm 1$;
- the tensor sound cone $\mu = -1/3$ (for $|k_1| = |k_2|$). On this shell the second-order vector response vanishes identically (Prop. 7), so the resonance belongs to the tensor sector alone.

The removable $\ln|\alpha|$ terms that Ci would introduce never appear. The complete expressions for ν_2, h_S, w and h are in `solution/results/second_order_closed_forms.txt`. Each antiderivative is verified by exact differentiation. Substituting the assembled ν_2 back into the lapse equation (N) gives an exact zero in closed form. The scalar part h_S satisfies (Ham) by construction.

The momentum and trace-free equations follow from three facts: the verified antiderivatives, the verified Green functions, and Thm. 5. They are also confirmed independently by the exact series test through $O(t^8)$. A direct closed-form substitution into (M) and (T) is available (`--check all`) but was not run to completion: it takes hours, and it is redundant with these checks.

Proposition 7 (Two-amplitude form on the shell)**.** *For any pair (k_1, k_2), reflection through the plane of the two momenta leaves a single TT polarisation. If moreover $|k_1| = |k_2|$ and $K = k_1 + k_2 \neq 0$, then $P_\perp c_M^{(2)} = 0$ identically in t, so the second-order vector response vanishes. With $u = \hat{K}$, $v = (k_1 - k_2)/|k_1 - k_2|$ (orthogonal to u on the shell) and $w = u \times v$,*

$$\boxed{A_{2,ij} = F_S(t, k_0, \mu)\Big(u_i u_j - \tfrac{1}{3}\delta_{ij}\Big) + F_T(t, k_0, \mu)\,(v_i v_j - w_i w_j), \qquad -1 < \mu < 1,} \tag{16}$$

with $F_S = h_S$ and F_T the single TT amplitude. The zero-total-momentum pair ($\mu = -1$) is the separately constructed $K = 0$ kernel, and the collinear pair ($\mu = 1$) has no v.

Proof. A parity-even quadratic vector source has the form $c_M = A\,k_1 + B\,k_2$, with $A(k_1, k_2) = B(k_2, k_1)$ by exchange symmetry. On the shell the scalar invariants are exchange-symmetric, so $A = B$ and $c_M \parallel K$. With no vector seed, $w = \frac{2i}{K^2}\int P_\perp c_M = 0$. For the tensor, the reflection $w \to -w$ removes the $vw + wv$ polarisation, and exchange ($v \to -v$) is compatible with $vv - ww$. `solution/check_shell_reduction.py` verifies $c_{M,z} = c_{T,xz} = c_{T,yz} = 0$ for general momenta, and $K \times c_M = 0$ on the shell identically in t and μ, with an off-shell negative control. □

This is an exact finite-time reduction. On the shell, the second-order metric is two scalar functions of (t, μ) instead of a component matrix.

Third order [explicit nested integrals]. For $C^{(13)}$ only the configuration $(k, q, -q)$ is needed. The source $c^{(3)}$ is explicit (`solution/results/third_order_sources.txt`) in the first-order closed forms and in $K_2(k, q)$, $K_2(k, -q)$ and $K_2(q, -q)$. The last of these is the exact $K = 0$ kernel, never a limit. The third-order kernels are the formulas above applied to $c^{(3)}$, so they are double integrals with closed-form inner integrals. Three levels should be distinguished:

- the general cubic response at arbitrary (k_1, k_2, k_3), which is *defined* constructively by the triangular scheme and the multilinear rule (Thm. 5 covers it);
- the configuration $(k, q, -q)$, which is the only one generated explicitly, because it is the only one $C^{(13)}$ needs;
- the checks of that configuration: an exact series test of all field equations through $O(t^6)$, and a finite-k numerical Ward test (App. A). Neither is a substitution of the assembled cubic kernel at arbitrary times.

6 The observable through third order

Proposition 8 (Tetrad)**.** *Fermi–Walker transport along u is parallel transport along ∂_t in $\tilde{g}$, that is, $\mathrm{d}E_a{}^i/\mathrm{d}t = -S^i{}_j E_a{}^j$. With $E(t_I) = \gamma(t_I)^{-1/2}$, the transported frame satisfies $E = \gamma_o^{-1/2} R$ at observation, where $R = 1 + \epsilon^2\Omega/2 + O(\epsilon^3)$ and Ω is antisymmetric. Consequently:*

- *$\Theta^{(1)}$ and $\Theta^{(2)}$ are those of the symmetric frame;*
- *the specified Fermi–Walker map is completed by*

$$\Theta^{(3)}_{\mathrm{FW}} = \Theta^{(3)}_{\mathrm{sym}} + 3(\Omega\hat{n})\cdot\nabla_{S^2}\Theta^{(1)}, \qquad \Omega = \frac{1}{4}\int_{\bar{t}_I}^{\bar{t}_o} [H_1(t,0), H_1'(t,0)]\,\mathrm{d}t; \tag{17}$$

- *$C_\ell^{(22)}$ and $\mathrm{Re}\, C_\ell^{(13)}$ are individually unchanged.*

Proof. Let $F = \gamma^{-1/2} = e^{-\zeta}e^{-H/2}$. From $E = FR$ and $E' = -SE$, $R' = -F^{-1}(SF + F')R$. Expanding the matrix exponentials gives $R' = \epsilon^2[H_1, H_1']/8 + O(\epsilon^3)$, with $R(t_I) = 1$. This proves the integral in (17); replacing its endpoints by their background values changes R only at $O(\epsilon^3)$, beyond the map order at issue. Since $\Theta_{\mathrm{FW}}(\hat{n}) = \Theta_{\mathrm{sym}}(R\hat{n})$, the Taylor convention gives the factor 3. The rotation acts on $a^{(1)}_{\ell m}$ by an anti-Hermitian generator, so $\mathrm{Re}\sum_m a^{(1)*}_{\ell m}(D(\Omega)a^{(1)})_{\ell m} = 0$ in every realisation. □

For a monochromatic seed, $H_1(t,0) = f(t)B$ with a time-independent matrix B, and hence $[H_1, H_1'] = 0$. In that specialisation even the maps agree through third order. The following ray and map formulas use the symmetric frame; (17) completes the map for general spectra after the clock composition.

Take the background ray $\bar{X} = \hat{n}(t_o - t)$, $\bar{e} = -\hat{n}$, and write $X = \bar{X} + \epsilon X_1 + \epsilon^2 X_2/2$. All fields are evaluated on $(t, \bar{X})$, and $\sigma(a,b) = \sigma_{ij}a^i b^j$. Because σ, S and Γ vanish on the background, X_3 and e_3 never enter. The exact consequences of Lemma 2 are as follows.

Integrand.

$$\begin{aligned} F_1 &= \sigma_{1,nn}, \qquad F_2 = \sigma_{2,nn} + 2X_1\cdot\partial\sigma_{1,nn} - 4\sigma_1(n,e_1),\\ F_3 &= \sigma_{3,nn} + 3X_1\cdot\partial\sigma_{2,nn} - 6\sigma_2(n,e_1) + 3X_2\cdot\partial\sigma_{1,nn} + 3(X_1\cdot\partial)^2\sigma_{1,nn}\\ &\quad - 12(X_1\cdot\partial)\sigma_1(n,e_1) - 6\sigma_1(n,e_2) + 6\sigma_1(e_1,e_1). \end{aligned}$$

Ray. The displacements are

$$X_r = e_r(t_o)(t - t_o) + \int_t^{t_o}(t' - t)A_r(t')\mathrm{d}t',$$

with

$$\begin{aligned} A_1 &= 2S_1 n - \Gamma_1(n,n) - n\sigma_{1,nn},\\ A_2 &= [2S_2 n - \Gamma_2(n,n) - n\sigma_{2,nn}] + 2(X_1\cdot\partial)[2S_1 n - \Gamma_1(n,n) - n\sigma_{1,nn}]\\ &\quad + 2[-2S_1 e_1 + 2\Gamma_1(n,e_1) + e_1\sigma_{1,nn} + 2n\sigma_1(n,e_1)]. \end{aligned}$$

The initial directions are the Taylor coefficients of $e(t_o) = -e^{-\zeta}\exp(-H/2)\,\hat{n}$, evaluated at $(t_o, 0)$.

Emission and endpoint. Let φ_r denote ν along the ray, with $\varphi_1 = \nu_1$, $\varphi_2 = \nu_2 + 2X_1 \cdot \partial\nu_1$, and so on. Then

$$d_1 = -\bar{t}_\star \varphi_1, \qquad d_2 = \bar{t}_\star[u_1^2 - \varphi_2 - 2\varphi_1' d_1], \qquad u_1 = d_1/\bar{t}_\star,$$
$$I_1 = \int F_1, \qquad I_2 = \int F_2 - 2F_1 d_1, \qquad I_3 = \int F_3 - 3F_1 d_2 - 3F_2 d_1 - 3F_1' d_1^2.$$

All unlabelled integrals here run from $\bar{t}_\star$ to t_o; endpoint factors are evaluated at $\bar{t}_\star$. Primes on ray-evaluated quantities denote total derivatives along the background ray.

Normalisation in centred form. At fixed coordinate observation time, put $J_c = I - \langle I \rangle_\Omega$, with coefficients $J_{c,r} = I_r - \langle I_r \rangle_\Omega$, and define $\mathsf{P}_0 f = f - \langle f \rangle_\Omega$. A sky-independent shift of I cancels exactly, so

$$\begin{aligned} \Theta_c &= \frac{e^{-J_c}}{\langle e^{-J_c} \rangle_\Omega} - 1, \\ \Theta_c^{(1)} &= -J_{c,1}, \\ \Theta_c^{(2)} &= -J_{c,2} + \mathsf{P}_0(J_{c,1}^2), \\ \Theta_c^{(3)} &= -J_{c,3} + 3\mathsf{P}_0(J_{c,1} J_{c,2}) - \mathsf{P}_0(J_{c,1}^3) + 3J_{c,1}\langle J_{c,1}^2 \rangle_\Omega. \end{aligned} \tag{18}$$

This is exactly the expanded normalisation formula, with all random-monopole products retained. The dipole is still present inside every product. It is removed only after the complete map is constructed; this final projection does not change $C_{\ell \geq 2}$ and is not a deboost.

Clock. With $\omega(t) = \nu(t, 0)$, define separately

$$W_1(T) = \int_0^T t\, \omega_1(t)\, \mathrm{d}t, \qquad W_2(T) = \int_0^T t\, [\omega_2(t) + \omega_1(t)^2]\, \mathrm{d}t.$$

The initial-time shifts d_{I1}, d_{I2} are the emission-root formulas with $\bar{t}_\star \to \bar{t}_I$ and $\varphi \to \omega$. The observation-time shifts are

$$\begin{aligned} D_1 &= [\bar{t}_I d_{I1} - W_1(\bar{t}_o) + W_1(\bar{t}_I)]/\bar{t}_o, \\ D_2 &= \big[\bar{t}_I d_{I2} + d_{I1}^2 - W_2(\bar{t}_o) - 2\bar{t}_o \omega_1(\bar{t}_o) D_1 \\ &\qquad + W_2(\bar{t}_I) + 2\bar{t}_I \omega_1(\bar{t}_I) d_{I1} - D_1^2\big]/\bar{t}_o. \end{aligned}$$

Compose with $t_o = \bar{t}_o + \epsilon D_1 + \epsilon^2 D_2/2 + \cdots$:

$$\begin{aligned} \Theta_{\mathrm{sym}}^{(2)} &= \Theta_c^{(2)} + 2D_1 \partial_{t_o} \Theta_c^{(1)}, \\ \Theta_{\mathrm{sym}}^{(3)} &= \Theta_c^{(3)} + 3D_2 \partial_{t_o} \Theta_c^{(1)} + 3D_1 \partial_{t_o} \Theta_c^{(2)} + 3D_1^2 \partial_{t_o}^2 \Theta_c^{(1)}. \end{aligned}$$

All right-hand sides are evaluated at $t_o = \bar{t}_o$. Equivalently, compose I with this same random time first, centre the resulting exponent, and use (18) once. These are alternative procedures, not two clock corrections to be applied successively.

Fourier dictionary. For a single mode a, with phase $e^{i\lambda k \cdot \hat{n}}$ and $\lambda = t_o - t$, $\mu_a = \hat{k} \cdot \hat{n}$, and $H_{a,ij} = Q_a(t)(k_i k_j - \delta_{ij} k^2/3)$, where $Q_a = -6[2\sin x/x - \cos x - 1]/k^2$:

- $\sigma_{a,nn} = \frac{1}{2} Q_a' k^2 (\mu_a^2 - \frac{1}{3})$;

- $\Gamma_a(n,n) = i[2(k\cdot n)n - k] + \frac{i}{2}[2(k\cdot n)H_a n - kH_{a,nn}]$;
- $e_a(t_o) = (1 + \frac{1}{2}H_a(t_o))\hat{n}$;
- $d_a = -\bar{t}_\star \nu_a(\bar{t}_\star) e^{i\lambda_\star k\cdot\hat{n}}$.

The second-order shear is $\sigma_{ab} = \frac{1}{2}H'_{ab} + \frac{1}{4}(H_aH_b + H_bH_a)' + H'_a + H'_b$, and

$$I_{ab} = \int_{\bar{t}_\star}^{t_o} \mathrm{d}t \left\{\sigma_{ab,nn} + i(k_b\cdot X_a)\sigma_{b,nn} + i(k_a\cdot X_b)\sigma_{a,nn} - 2\sigma_b(n,e_a) - 2\sigma_a(n,e_b)\right\} - [F_a d_b + F_b d_a]_{\bar{t}_\star},$$
$$\Theta_{ab} = I_aI_b - I_{ab} - \langle I_a\rangle_\Omega I_b - \langle I_b\rangle_\Omega I_a + 2\langle I_a\rangle_\Omega\langle I_b\rangle_\Omega - \langle I_aI_b\rangle_\Omega + \langle I_{ab}\rangle_\Omega + D_a\partial\Theta_b + D_b\partial\Theta_a.$$

A compact expression for *all* map kernels is obtained by taking $\mathcal{J}_a$, $\mathcal{J}_{ab}$ and $\mathcal{J}_{abc}$ to be the centred-exponent kernels *after* clock composition. In the symmetric frame, (18) polarises to

$$\begin{aligned}
\mathcal{K}_1(a) &= -\mathcal{J}_a,\\
\mathcal{K}_2(a,b) &= -\mathcal{J}_{ab} + \mathsf{P}_0(\mathcal{J}_a\mathcal{J}_b),\\
\mathcal{K}_3(a,b,c) &= -\mathcal{J}_{abc} + \mathsf{P}_0\left(\sum_{\mathrm{cyc}} \mathcal{J}_a\mathcal{J}_{bc} - \mathcal{J}_a\mathcal{J}_b\mathcal{J}_c\right) + \sum_{\mathrm{cyc}} \mathcal{J}_a\langle\mathcal{J}_b\mathcal{J}_c\rangle_\Omega.
\end{aligned} \tag{19}$$

Each cyclic sum has three terms. The coefficient convention of §2 introduces no further factorials. For the specified general-spectrum map, add the multilinear coefficient of (17); for the spectra below the symmetric-frame kernels suffice. Thus the lengthy expanded kernels can remain in the ancillary files without obscuring the finite-time structure of the final observable.

Independent generator. `observable_generator.py` rebuilds Θ *from the exact definitions* by fixed-point iteration in the multilinear algebra: the ray ODE, the emission root, the clock equation and the sky normalisation. It uses no hand-expanded formula, and it passes the Ward identities and the superhorizon check of §7 and §9.1.

7 Gauge invariance and Ward identities

Every coordinate and label is a physical scalar or frame: t, X, the surface $T = T_\star$, the fluid line $X = 0$, proper time from $T = T_I$, and $\gamma^{-1/2}$ in the labels. So $\Theta(\hat{n})$ is invariant under any diffeomorphism acting on fields and labels together. The only residual freedom is the conformal group of the primordial chart (§4). Its dilation generator must act as $\zeta \to \zeta + c \Leftrightarrow k \to e^{-c}k$. We verify this exactly:

- *Fields:* $K_2(k,0) = -k\cdot\partial_k K_1$ and $K_3(k,0,0) = (k\cdot\partial_k)^2 K_1$ solve every equation in closed form. Perturbed predictions fail.
- *Observable:* $\Theta_{sL} = -\kappa\partial_\kappa\Theta_s$ and $\Theta_{sLL} = (\kappa\partial_\kappa)^2\Theta_s$ hold through $O(\kappa^4)$, where κ scales the momenta. At this order the ray deflection, time delay, emission shift, clock, frame and normalisation all contribute.
- At first order, the result equals the independent Newtonian-gauge computation (SW, Doppler and ISW) for every ℓ.

7.1 Gradient-mode tests and relation to the literature

Mirbabayi and Zaldarriaga, *CMB Anisotropies from a Gradient Mode* [1], provide two especially useful comparisons. Their §§3–4 show why a physical emission surface, the observer's position and velocity, the observer clock, and the measured temperature normalisation must be retained for a long-gradient cancellation. Their §5 distinguishes an uncompensated primordial gradient, which generates a second-order quadrupole, from a nonlinearly completed adiabatic gradient, which is locally a coordinate transformation. Appendix C demonstrates cancellation of inverse-Laplacian terms in the long-wavelength observable, including boundary contributions.

The symmetry arguments are relevant here, but their numerical transfer coefficients cannot be imported. Their explicit dynamics are matter dominated; the long-long calculation also neglects η_e/η_o, and the squeezed calculation neglects terms of order $(\lambda_S/\lambda_L)^2$. This report instead retains radiation dynamics and finite emission and observation times. The following checks are derived directly in the temperature frame, without making those matter-era approximations.

Proposition 9 (Exact adiabatic-gradient null test)**.** *Let b be a constant vector, and in a nonsingular patch containing the observer define*

$$D_b(X) = 1 - 2b\cdot X + b^2X^2, \qquad Y_b(X) = \frac{X - bX^2}{D_b(X)}, \qquad \zeta_{\rm ad}(X) = -\ln D_b(X). \tag{20}$$

Then the radiation solution $\nu = H = 0$, $\gamma = D_b^{-2}\delta$ has $\Theta(\hat n) = 0$ exactly, at arbitrary finite observation and emission times within the patch.

Proof. The special conformal transformation satisfies $(\partial_X Y_b)^T(\partial_X Y_b) = D_b^{-2}\mathbf{1}$. Thus γ is a flat metric written in different spatial labels, and $g = (a_1 t)^2(-\mathrm{d}t^2 + \mathrm{d}Y_b^2)$ is the unperturbed radiation spacetime. In particular $R_{ij}[\gamma] = 0$, $S = 0$ and all four exact field equations hold. The observer remains at $Y_b(0) = 0$, with $\partial_X Y_b|_0 = \mathbf{1}$, so the specified initial orientation and clock are unchanged. Since $\sigma = 0$, the exact redshift exponent is $I = 0$ for every ray. Normalisation gives $\Theta = 0$, including before dipole subtraction. □

With $b = \epsilon p/2$, the completed primordial profile is

$$\zeta_{\rm ad} = \epsilon\, p\cdot X + \epsilon^2\left[\frac{1}{2}(p\cdot X)^2 - \frac{1}{4}p^2X^2\right] + O(\epsilon^3). \tag{21}$$

The bracket is the coefficient of ϵ^2, not $\zeta^{(2)}$; the latter is twice the bracket in our convention. The completed seed has vanishing curvature, so it also passes (46). An explicit local check is $e^{-\zeta_{\rm ad}} = D_b$, whose Hessian is $2b^2\delta_{ij}$ and has zero trace-free part.

By contrast, for the *uncompleted* local profile $\zeta = \epsilon p\cdot X$ at $X_o = 0$, the leading-gradient formula gives

$$\Theta_{\rm grad,lead}(\hat n) = \epsilon^2\tau^2\left[(p\cdot\hat n)^2 - \frac{1}{3}p^2\right]. \tag{22}$$

It vanishes at first order but not at second order. To compare the primordial completion with [1], use *their* matter-era notation $\phi_1 = q\cdot X = -3\zeta_1/5$, hence $p = -5q/3$. Multiplying the quadratic bracket in (21) by $-3/5$ gives exactly their $\phi_2 = 5q^2X^2/12 - 5(q\cdot X)^2/6$. This agreement concerns the adiabatic initial profile, not equality of the matter and radiation quadrupole coefficients. In particular, adding that quadratic profile would change the Gaussian initial-data prescription of Def. 4; it must not be used to cancel the physical contribution in this report.

Proposition 10 (Soft-gradient identity for the finite-time map)**.** *For the kernels of the complete map at fixed physical labels, assuming differentiability in the soft momentum at zero,*

$$\boxed{\partial_{q_i}\mathcal{K}_2(k,q;\hat{n})|_{q=0} = -\mathcal{D}_i\mathcal{K}_1(k;\hat{n}), \qquad \mathcal{D}_i = k_j\partial_{k_i}\partial_{k_j} - \frac{1}{2}k_i\Delta_k.} \tag{23}$$

Momentum derivatives keep $\hat{n}$, $T_\star$, T_I and τ_o fixed. This identity is valid at general short-mode momentum and finite times; it is not restricted to the superhorizon expansion of the short mode.

Proof. For an arbitrary regular scalar seed ζ_s, its pullback by (20) is the seed $\zeta_s(Y_b(X)) - \ln D_b(X)$. The same observer and tetrad are selected at the origin. Diffeomorphism invariance therefore makes the complete measured map identical for these two seed descriptions. To first order in an auxiliary parameter λ, set $b = \lambda p/2$ and write

$$Y_b = X + \lambda\xi_p(X) + O(\lambda^2), \qquad \xi_p(X) = (p\cdot X)X - \tfrac{1}{2}X^2 p.$$

For a short seed $sh(X)$ the pulled-back seed is $sh + \lambda p\cdot X + s\lambda\xi_p\cdot\nabla h + \cdots$. If $\mathcal{L}$ and $\mathcal{B}$ denote the linear and symmetric bilinear map responses (with the Taylor prefactor $1/2$ on $\mathcal{B}$), equality of the $s\lambda$ coefficients gives

$$\mathcal{B}[h, p\cdot X] = -\mathcal{L}[\xi_p\cdot\nabla h].$$

For $h = e^{ik\cdot X}$, the left-hand side is $-ip_i\partial_{q_i}\mathcal{K}_2|_{q=0}$. The initial deformation on the right is $\xi_p\cdot\nabla e^{ik\cdot X} = -ip_i\mathcal{D}_i e^{ik\cdot X}$, with the displayed k_j coefficients outside the derivatives. Linearity then gives (23). □

Combining this with the dilation identity isolates a soft remainder:

$$\mathcal{R}_2(k,q;\hat{n}) \equiv \mathcal{K}_2(k,q;\hat{n}) + k\cdot\partial_k\mathcal{K}_1(k;\hat{n}) + q_i\mathcal{D}_i\mathcal{K}_1(k;\hat{n}) = O(q^2). \tag{24}$$

Thus both the constant and gradient parts of a soft quadratic response are fixed by the linear transfer function; genuinely new soft information starts at two gradients. This is a useful organisation and consistency condition for the large ancillary kernels. The symmetry proof is exact, and the leading-gradient quadratic kernel independently satisfies (23); a direct arbitrary-time substitution of the full assembled quadratic observable into this identity has not been performed.

What this does and does not cancel. The no-modulation result for a gradient in [1] includes the appropriate transformation of short-mode initial data (or the corresponding inflationary consistency condition). An exactly Gaussian g with an arbitrary spectrum in a fixed marked chart does not by itself impose that condition. Equations (23) and (24) therefore do not license dropping the soft pieces of $\mathcal{K}_2$, or the gradient-squared term in (22), from this problem's spectrum. Nor do long-wavelength cancellations prove that the kernels at comparable, finite momenta reduce to local curvature polynomials. In the notation of §9.1, the positive m_2^2 contribution is retained.

7.2 Further literature

The following work bears directly on particular steps. None of it is used as an input: every result above is derived independently in the temperature frame. Each item states what it corroborates and what has *not* been compared.

- *Adiabatic modes and their Ward identities* [2, 3, 4]. The dilation and special-conformal identities of §7 and Prop. 10 are the lowest members of this tower. The dilation factorisation (49) is the Gaussian average of the first member.

- *Local observers and curvature* [5, 6]. Conformal Fermi coordinates make long modes appear locally only through curvature. This matches the seed-Ricci form (46) and the statement that only the trace-free curvature quadrupole survives at leading gradient order.
- *Superhorizon modes and the CMB quadrupole* [7, 8]. The $C_2 \propto \Xi^2 \propto k_0^4$ behaviour of §9.1 is the Grishchuk–Zel'dovich effect, here in the radiation era, with finite emission time and exactly in ζ.
- *Nonlinear Sachs–Wolfe effect* [9, 10]. These papers show that the second-order large-scale temperature is far simpler than its metric ingredients. Lemma 2 ($T_{\rm obs} \propto e^{-I}$ exactly) is the temperature-frame version of that simplification. Their matter-era coefficients are not comparable.
- *Nonlinear ζ and the gradient expansion* [11, 12, 13]. These support Def. 4: they give the conservation of $\zeta = \frac{1}{6}\ln\det\gamma$ and the uniform-density (δN) interpretation, and (5) is the radiation-era leading term of their expansion.
- *Scalar-induced tensors in the radiation era* [14, 15, 16]. These give closed-form Si/Ci kernels for the second-order tensor sourced by two first-order scalars, including the monochromatic spectrum and its resonance at $K = 2k_0/\sqrt{3}$. That resonance is the sound-cone surface $\mu = -1/3$ of §5, where the entire-function kernels are manifestly finite. A term-by-term comparison of our TT kernel h with theirs (which needs a gauge map from Newtonian to temperature-frame variables) has *not* been made. It is the most direct external test available for the second-order field solution.

8 The spectrum

8.1 Gaussian reduction

With $\mathcal{K}_r(k_1..k_r;\hat{n})$ the kernel of $\Theta^{(r)}$ (observer at the origin) and $A^{(r)}_{\ell m} = \int \mathrm{d}\Omega\, Y^*_{\ell m}\mathcal{K}_r$:

$$C_\ell^{(11)} = \tfrac{1}{2\ell+1}\textstyle\sum_m \int_k P|A^{(1)}_{\ell m}|^2, \qquad C_\ell^{(22)} = \tfrac{2}{2\ell+1}\textstyle\sum_m \int_k\int_q P(k)P(q)|A^{(2)}_{\ell m}(k,q)|^2,$$
$$C_\ell^{(13)} = \tfrac{3}{2\ell+1}\textstyle\sum_m \int_k\int_q P(k)P(q)A^{(1)}_{\ell m}(k)A^{(3)}_{\ell m}(k,q,-q)^*.$$

The disconnected part of 22 is proportional to $\delta_{\ell 0}$. In 13 all three pairings survive; the internal $(q,-q)$ pairing is a pairing of the fourth moment and is not removed by normal ordering. Odd orders vanish.

Compact finite-time spectrum. For $\ell \geq 2$, define the corrected linear response

$$A^{\rm eff}_{\ell m}(k) = A^{(1)}_{\ell m}(k) + \frac{\epsilon^2}{2}\int_q P(q)A^{(3)}_{\ell m}(k,q,-q). \tag{25}$$

Then the complete result through fourth order can be written as two norms:

$$\boxed{C_\ell = \frac{1}{2\ell+1}\sum_m\left\{\epsilon^2\int_k P(k)|A^{\rm eff}_{\ell m}(k)|^2 + \frac{\epsilon^4}{2}\int_k\int_q P(k)P(q)|A^{(2)}_{\ell m}(k,q)|^2\right\} + O(\epsilon^6).} \tag{26}$$

The square is understood only through $O(\epsilon^4)$: its $O(\epsilon^6)$ part is not a prediction. To see the factor $1/2$ in (25), Wick-decompose the cubic Gaussian product into its normal-ordered part plus its

three single contractions. The cubic prefactor $1/6$ turns these into a $1/2$ correction to the linear response; the remaining normal-ordered cubic part is orthogonal to the linear part. The quadratic contribution has two connected pairings and its prefactor is $(1/2)^2$, giving the other $1/2$ in (26). No 13 interference term has been discarded or absorbed into a redefinition of the primordial variance. The formula applies whenever the displayed integrals exist, in particular to the monochromatic case.

Equations (19)–(26), together with the factored field sources in §5.2, are a compact organisation of the *general finite-time* answer. They retain the same integrals as the expanded result; they do not establish an additional cancellation of the cubic response or evaluate its momentum and time integrals.

Linear transfer function. For $\ell \geq 2$, the convention is

$$\begin{aligned} A^{(1)}_{\ell m}(k) &= 4\pi i^\ell T_\ell(k) Y^*_{\ell m}(\hat{k}), \\ T_\ell(k) &= -3\int_{x_\star}^{x_o} s'(x)\, w_\ell\Big(\sqrt{3}(x_o - x)\Big)\, \mathrm{d}x, \\ s(x) &= \frac{2\sin x}{x} - \cos x, \qquad w_\ell(y) = \frac{1}{3} j_\ell(y) + j''_\ell(y). \end{aligned} \tag{27}$$

The prime on s differentiates x, and those on j_ℓ differentiate its full argument. Thus $C_\ell^{(11)} = 4\pi \int \mathcal{P}(k) T_\ell(k)^2 \,\mathrm{d}\ln k$.

Closed form of T_ℓ for every ℓ [closed form]. The integrand is a product of a sound wave in the acoustic phase x and a spherical wave in the photon phase $\sqrt{3}(x_o - x) = k(t_o - t)$. Split both into their Hankel (outgoing) parts,

$$\begin{aligned} s'(x) &= \mathrm{Re}\left[e^{ix} S(x)\right], \quad S(x) = -i + \frac{2}{x} + \frac{2i}{x^2}, \\ w_\ell(y) &= \mathrm{Re}\left[e^{iy} V_\ell(y)\right], \quad V_\ell(y) = e^{-iy}\Big(\frac{1}{3} + \partial_y^2\Big) h^{(1)}_\ell(y), \end{aligned} \tag{28}$$

and put $V^+_\ell = V_\ell$, $V^-_\ell = \overline{V_\ell}$ (conjugate coefficients), $\omega_\sigma = 1 - \sigma\sqrt{3}$ and $\Delta = x_o - x_\star$. Both S and V_ℓ are Laurent polynomials; for example

$$\begin{aligned} V_2(y) &= -\frac{2i\,(y^4 + 6iy^3 - 24y^2 - 54iy + 54)}{3y^5}, \\ V_3(y) &= -\frac{2\,(y^5 + 9iy^4 - 54y^3 - 204iy^2 + 450y + 450i)}{3y^6}. \end{aligned}$$

Proposition 11 (Entire-function form of the transfer function)**.** *Use the subtracted kernels of §5,*

$$J_m(\omega, t) = \int_0^t s^{-m}\Big[e^{i\omega s} - \sum_{j<m} \frac{(i\omega s)^j}{j!}\Big]\mathrm{d}s,$$

each of which is t^{1-m} times an entire function of ωt. Then

$$\boxed{\begin{aligned} T_\ell = -\frac{3}{2}\,\mathrm{Re}\sum_{\sigma=\pm1}\Big\{ & e^{i\sigma\sqrt{3}x_o}\sum_{m=1}^{2} a^\sigma_m \big[J_m(\omega_\sigma, x_o) - J_m(\omega_\sigma, x_\star)\big] \\ & + e^{ix_o}\sum_{m=1}^{\ell+3} b^\sigma_m\, J_m(-\omega_\sigma, \Delta)\Big\}, \end{aligned}} \tag{29}$$

where the coefficients depend on x_o only:

$$a_2^\sigma = 2iV_\ell^\sigma(\sqrt{3}x_o), \quad a_1^\sigma = 2V_\ell^\sigma(\sqrt{3}x_o) - 2\sqrt{3}\,i\,V_\ell^{\sigma\prime}(\sqrt{3}x_o), \quad b_m^\sigma = \left[u^{-m}\right] S(x_o-u)\,V_\ell^\sigma(\sqrt{3}u),$$

with $[u^{-m}]$ denoting the Laurent coefficient at $u=0$.

Proof. By (28), $-3s'w_\ell = -\frac{3}{2}\operatorname{Re}\sum_\sigma e^{i\sigma\sqrt{3}x_o}e^{i\omega_\sigma x}R_\sigma(x)$, where $R_\sigma(x) = S(x)V_\ell^\sigma(\sqrt{3}(x_o-x))$ is rational and vanishes at infinity. Its only poles are at $x=0$ (order two, from S) and $x=x_o$ (order $\ell+3$, from V_ℓ), so R_σ is the sum of its two principal parts. These are $a_2^\sigma/x^2 + a_1^\sigma/x$ and, in $u = x_o - x$, $\sum_m b_m^\sigma u^{-m}$. Write each $e^{i\omega x}x^{-m}$ as the subtracted integrand of J_m plus the principal part of $e^{i\omega x}x^{-m}$. The discarded pieces add up to the principal part, at $x=0$ and at $x=x_o$, of the complex integrand. The real part of that sum is the principal part of $-3s'(x)w_\ell(\sqrt{3}(x_o-x))$, which is regular at both points ($s'(x) = x/3 + O(x^3)$, and w_ℓ is entire). Hence the discarded pieces vanish under Re, and term-by-term integration gives (29). □

The representation has no logarithms, no Ci and no elementary remainder. It uses the same entire functions as the second-order kernels, and the only frequencies are the beats $\sqrt{3}\mp 1$ between the photon and acoustic phases. Every term is manifestly finite as $x_\star \to x_o$ and as $x_\star \to 0$. If preferred, the recursion for J_M reduces everything to $J_1 = -\operatorname{Cin} + i\operatorname{Si}$ together with elementary terms. Expressed instead in Ci and Si, the same function carries removable $\ln(\sqrt{3}\pm 1)$ constants. `solution/transfer_entire.py` proves (29) exactly for $\ell = 1,\dots,6$, as an identity in $x_\star$ together with its value at $x_\star = x_o$. The case $\ell = 1$, the dipole of $\Theta^{(1)}$, is needed in §8.2.

8.2 Log-map decomposition of the fourth-order spectrum

By Lemma 2, the centred *log-temperature map*

$$L(\hat n) \equiv \ln T_{\rm obs}(\hat n) - \langle \ln T_{\rm obs}\rangle_\Omega = -\big(I(\hat n) - \langle I\rangle_\Omega\big) \tag{30}$$

is an observable. It is linear in the redshift exponent I and contains no normalisation nonlinearity; the clock composition and the frame of §6 are included. The map is recovered from it exactly by $\Theta = e^L/\langle e^L\rangle_\Omega - 1$. Write $L = \epsilon\theta + \epsilon^2 L_2/2 + \epsilon^3 L_3/6$. Then $\theta = \Theta^{(1)}$ is Gaussian and isotropic, with spectrum $C_\ell \equiv C_\ell^{(11)}$ and correlation function $\xi(\hat n\cdot\hat n') = \sum_{\ell\ge1}\frac{2\ell+1}{4\pi}C_\ell P_\ell(\hat n\cdot\hat n')$.

Theorem 12 (Log-map decomposition)**.** *For $\ell \ge 1$, exactly,*

$$\boxed{\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\operatorname{Re}C_\ell^{(13)} = C_\ell^{L,4} + \mathcal{N}_\ell[C] + \mathcal{M}_\ell,} \tag{31}$$

where, writing $C_\ell[F,G] = \frac{1}{2\ell+1}\sum_m\langle F_{\ell m}G^*_{\ell m}\rangle$, the three terms are the fourth-order spectrum of the log map, the lognormal normalisation and the collapsed log-map bispectrum:

$$C_\ell^{L,4} = \tfrac{1}{4}C_\ell[L_2,L_2] + \tfrac{1}{3}\operatorname{Re}C_\ell[\theta,L_3], \tag{32}$$

$$\mathcal{N}_\ell[C] = \frac{1}{2}\sum_{\ell_1,\ell_2\ge1}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell_1 & \ell_2 & \ell\\ 0 & 0 & 0\end{pmatrix}^2 C_{\ell_1}C_{\ell_2} - \frac{C_\ell^2}{2\pi}, \tag{33}$$

$$\mathcal{M}_\ell = \tfrac{1}{2}\operatorname{Re}C_\ell[L_2,\theta^2] + \operatorname{Re}C_\ell[\theta,\theta L_2]. \tag{34}$$

In real space, with $w^{(4)}$ the ϵ^4 coefficient of $\langle\Theta(\hat n)\Theta(\hat n')\rangle$ (for $\ell\ge1$) and $(\xi\star\xi)(\hat n\cdot\hat n') = \int \mathrm{d}\Omega_m\,\xi(\hat n\cdot\hat m)\,\xi(\hat m\cdot\hat n')$,

$$w^{(4)} = \tfrac{1}{4}\langle L_2L_2'\rangle + \tfrac{1}{3}\langle\theta L_3'\rangle + \tfrac{1}{2}\xi^2 - \tfrac{1}{2\pi}\,\xi\star\xi + \tfrac{1}{2}\langle\theta^2L_2'\rangle + \langle\theta\,\theta' L_2'\rangle, \tag{35}$$

where primes denote evaluation at $\hat n'$, and the right-hand side is symmetrised in $\hat n \leftrightarrow \hat n'$.

Proof. With $J_c = -L$, (18) gives $\Theta^{(2)} = L_2 + \mathsf{P}_0\theta^2$ and $\Theta^{(3)} = L_3 + 3\mathsf{P}_0(\theta L_2) + \mathsf{P}_0\theta^3 - 3\theta\langle\theta^2\rangle_\Omega$; the projections P_0 affect only $\ell = 0$. Insert these into the bilinear spectra and collect terms. For the terms in θ alone, Wick's theorem gives $\langle\theta^2\theta'^2\rangle_c = 2\xi^2$, $\langle\theta\,\theta'^3\rangle = 3\xi\,\xi(1)$ and $\langle\theta\theta'\langle\theta^2\rangle_\Omega\rangle = \xi\,\xi(1) + \frac{2}{4\pi}\xi\star\xi$. The variance terms $\xi(1)$ cancel between the last two. The harmonic form follows from $P_{\ell_1}P_{\ell_2} = \sum_L (2L+1)\left(\begin{smallmatrix}\ell_1 & \ell_2 & L\\ 0 & 0 & 0\end{smallmatrix}\right)^2 P_L$ and from the fact that $\xi\star\xi$ has Legendre coefficients C_ℓ^2. □

Three features make (31) the natural organisation of the answer.

1. *The lognormal term is explicit.* $\mathcal{N}_\ell$ depends on $C^{(11)}$ alone. By Prop. 11 it is an explicit convergent harmonic series for every ℓ; it is not a finite closed expression. Equivalently, it is a single angular integral that we do not evaluate:

$$\mathcal{N}_\ell = \pi\int_{-1}^{1} P_\ell(u)\,\xi(u)^2\,\mathrm{d}u - \frac{C_\ell^2}{2\pi}, \qquad \ell \geq 1. \tag{36}$$

   The map variance $\xi(1)$ cancels identically. The dipole ($\ell_1 = 1$ or $\ell_2 = 1$) enters explicitly, because it is removed only after the nonlinear normalisation, as the problem requires. The term $-C_\ell^2/2\pi$ is the effect of normalising by the *random* sky monopole. On the monochromatic shell,

$$\boxed{\mathcal{N}_\ell = 2\pi\sigma^4\Big[\sum_{\ell_1,\ell_2\geq 1}(2\ell_1+1)(2\ell_2+1)\begin{pmatrix}\ell_1 & \ell_2 & \ell\\ 0 & 0 & 0\end{pmatrix}^2 T_{\ell_1}^2 T_{\ell_2}^2 - 4T_\ell^4\Big].} \tag{37}$$

   The sum converges absolutely. T_ℓ is an integral of $w_\ell(y)$ over $y \in [0, \sqrt{3}(x_o - x_\star)]$, so it decays faster than any power once ℓ exceeds the maximal argument.

2. *The mixed term needs only a three-point function.* $\mathcal{M}_\ell$ involves only the leading bispectrum $\langle\theta\theta L_2\rangle$ of the log map, in two collapsed configurations. L_3 does not enter it.

3. *All cubic response sits in the log-map spectrum.* L_3 appears only in $C_\ell^{L,4}$. The quadratic dynamics (L_2) enters both $C_\ell^{L,4}$ and $\mathcal{M}_\ell$. $\mathcal{N}_\ell$ is pure measurement nonlinearity (exponentiation and normalisation). $\mathcal{M}_\ell$ is the interference of that nonlinearity with the quadratic dynamics.

4. *Gradient counting.* In the superhorizon limit $\theta = O(\kappa^2)$, so $\mathcal{N}_\ell = O(\kappa^8)$. Power counting alone gives $\mathcal{M}_\ell = O(\kappa^6)$, but that order cancels. At leading gradient order $\theta = -\tau^2 A_{nn}$ and $L_2 = \tau^2(4gA + 2B)_{nn}$, with $A = (\partial\partial g)_{\mathrm{TF}}$ and $B = (\partial g\,\partial g)_{\mathrm{TF}}$. At a point, g, A and ∇g are independent, and $\langle g\rangle = \langle B\rangle = 0$. Hence $\langle\theta(\hat{n})\theta(\hat{n}')L_2(\hat{m})\rangle = 0$ for arbitrary directions at $O(\kappa^6)$; spatial parity removes $O(\kappa^7)$. Therefore

$$\mathcal{M}_\ell = O(\kappa^8), \qquad \mathcal{N}_\ell = O(\kappa^8),$$

   and the fourth-order temperature and log-temperature spectra agree through total gradient order six. This order contains the leading quadrupole ($O(k_0^4)$, §9.1) and the leading octupole ($O(k_0^6)$, §9.2).

`solution/check_logmap_decomposition.py` verifies (31) and (33) exactly for $\ell = 1, \ldots, 4$. It uses a band-limited isotropic Gaussian sky ($\ell = 1, 2$ with independent spectra) and an independent dipole field χ, with non-Gaussian $L_2 = \mathsf{P}_0(\theta\chi + \theta^2)$ and $L_3 = \mathsf{P}_0(\theta\chi^2)$. The map is expanded from its exact definition, and every average is an exact sphere integral and Gaussian moment. For a pure quadrupole, for example, $\mathcal{N}_2 = -\frac{9}{28\pi}C_2^2$.

8.3 Observer-centred dilation decomposition

Split the primordial field into its value at the observer and the observer-centred remainder, $g = g_o + g_c$, with $g_o = g(0)$ and $g_c = g - g(0)$. A constant shift of ζ is exactly a momentum dilation (§7). Hence $\Theta[\epsilon g] = \exp(-\epsilon g_o \mathfrak{D})\,\Theta[\epsilon g_c]$ exactly, where $\mathfrak{D} = \sum_a k_a \cdot \partial_{k_a}$ acts on the kernels at fixed $\hat{n}$ and fixed physical labels. Expanding gives the kernels of the observer-centred field:

$$\begin{aligned}\mathcal{K}_2^c(k,q) &= \mathcal{K}_2(k,q) + \mathfrak{D}\mathcal{K}_1(k) + \mathfrak{D}\mathcal{K}_1(q),\\ \mathcal{K}_3^c(k,q,-q) &= \mathcal{K}_3(k,q,-q) + \mathfrak{D}\big[\mathcal{K}_2^c(k,q) + \mathcal{K}_2^c(k,-q) + \mathcal{K}_2^c(q,-q)\big]\\ &\quad - \mathfrak{D}^2\big[\mathcal{K}_1(k) + \mathcal{K}_1(q) + \mathcal{K}_1(-q)\big].\end{aligned}$$

Write $A^c_{r,\ell m}$ for their harmonic coefficients and $D = k\partial_k$.

Proposition 13 (Observer-dilation decomposition)**.** *For $\ell \geq 1$ and any spectrum for which the integrals exist, with $\sigma^2 = \langle g^2\rangle = \int_k P$,*

$$\boxed{\begin{aligned}\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} &= C_\ell^{c,4} + \frac{\sigma^2}{2(2\ell+1)}\int_k P(k)\,D^2\sum_m |A^{(1)}_{\ell m}(k)|^2\\ &\quad - \frac{2}{2\ell+1}\int_k\!\!\int_q P(k)P(q)\,\mathfrak{D}\,\mathrm{Re}\sum_m A^c_{2,\ell m}(k,q)A^{(1)*}_{\ell m}(k).\end{aligned}} \tag{38}$$

Here $C_\ell^{c,4}$ is the fourth-order spectrum built from the observer-centred kernels,

$$C_\ell^{c,4} = \frac{1}{2(2\ell+1)}\sum_m \int\!\!\int PP\,|A_2^c|^2 + \frac{1}{2\ell+1}\,\mathrm{Re}\sum_m \int\!\!\int PP\,A^{(1)}A_3^{c*}.$$

On the monochromatic shell, with σ^2 held fixed, the last two terms become $\frac{1}{2}\sigma^2 D_{k_0}^2 C_\ell^{(11)}$ and a total derivative $-2D_{k_0}X_\ell$, where $D_{k_0} = k_0\,\mathrm{d}/\mathrm{d}k_0$ at fixed times and $X_\ell = \frac{1}{2\ell+1}\,\mathrm{Re}\sum_m \iint PP\,A_2^c A^{(1)*}$.

Proof. Insert the kernel relations into the Gaussian reductions of §8. Several terms are independent of $\hat{k}$ and vanish against $A^{(1)}_{\ell m}(k)$ after the $\hat{k}$ integral for $\ell \geq 1$: $\mathfrak{D}\mathcal{K}_2^c(q,-q)$ and $\mathfrak{D}^2\mathcal{K}_1(\pm q)$ in $\mathcal{K}_3$, and the cross term $\mathfrak{D}\mathcal{K}_1(k)\,\mathfrak{D}\mathcal{K}_1(q)^*$ in $|\mathcal{K}_2|^2$. The purely dilation parts give $\sigma^2\int P\,[\,|DA^{(1)}|^2 + \mathrm{Re}\,A^{(1)}D^2A^{(1)*}] = \frac{\sigma^2}{2}\int P\,D^2|A^{(1)}|^2$. In the dilation–centred interference, the 22 term contributes $-2\,\mathrm{Re}\,A_2^c\,DA^{(1)*}$ (by $k \leftrightarrow q$ symmetry) and the 13 term contributes $-2\,\mathrm{Re}\,A^{(1)}(\mathfrak{D}A_2^c)^*$ (by $q \to -q$). Their sum is the total dilation derivative $-2\,\mathfrak{D}\,\mathrm{Re}\,A_2^cA^{(1)*}$, because $A^{(1)}(k)$ does not depend on q. On the shell, $\iint PP\,\mathfrak{D}F = D_{k_0}\iint PP\,F$ at fixed σ^2. □

Three remarks place this result.

- *Relation to the infrared theorem.* Integrating by parts in $\ln k$, the dilation term equals $\frac{\sigma^2}{2(2\ell+1)}\int_k \sum_m |A^{(1)}_{\ell m}|^2(D+3)^2P$. This is Thm. 22 with the long-mode variance σ_L^2 replaced by the full variance. The observer's own value of the field acts as a long mode for every wavelength.

- *Superhorizon limit.* At leading gradient order, $\mathcal{K}_1 = \tau^2\,\mathrm{TF}(kk)_{nn}$, and two exact simplifications follow: $\mathcal{K}_2^c = -\tau^2\,\mathrm{TF}(kq+qk)_{nn}$ is pure gradient-squared, and $\mathcal{K}_3^c(k,q,-q) = 0$ identically. The cross term vanishes after the $\hat{q}$ average. Hence $\frac{29}{3} = 8 + \frac{5}{3} + 0 + 0$: the dilation term is exactly the 8, the centred 22 part is the $\frac{5}{3}$, and the centred 13 part and the cross term vanish. The decomposition (38) is the finite-k_0 extension of (49).

- *Status.* No analytical statement is made here about the relative size of the three terms at finite k_0. In the sampled configurations of the exploratory appendix (App. A, not evidence), the dilation term is not dominant. Whether the centred part or the cross term carries simple structure is open.

8.4 Monochromatic spectrum

After the complete observable is constructed, set

$$\mathcal{P}(k) = \sigma^2 \delta(\ln k/k_0), \qquad P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\delta(k-k_0), \qquad \langle g^2\rangle = \sigma^2 \text{ held fixed.}$$

The background and the labels $T_\star$, T_I and τ_o are also held fixed. Using joint rotation invariance, with $\hat{k} = \hat{z}$ and $\hat{q}_\mu = (\sqrt{1-\mu^2}, 0, \mu)$:

$$C_\ell^{(11)} = 4\pi\sigma^2 T_\ell(k_0)^2, \tag{39}$$

$$C_\ell^{(22)} = \frac{\sigma^4}{4\pi}\int_{-1}^{1}\mathrm{d}\mu \int \mathrm{d}\Omega \int \mathrm{d}\Omega'\, P_\ell(\hat{n}\cdot\hat{n}')\, \mathcal{K}_2(\hat{n})\mathcal{K}_2(\hat{n}')^*, \tag{40}$$

$$C_\ell^{(13)} = \frac{3\sigma^4}{2}\, i^\ell T_\ell(k_0) \int_{-1}^{1}\mathrm{d}\mu \Big[\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\, \mathcal{K}_3(k_0\hat{z}, k_0\hat{q}_\mu, -k_0\hat{q}_\mu; \hat{n})\Big]^*. \tag{41}$$

Primary form of the answer. Put $v = \epsilon^2\sigma^2$ and define two dimensionless response functions of k_0 at fixed times:

$$Q_\ell = \frac{1}{4(2\ell+1)}\int_{-1}^{1}\mathrm{d}\mu \sum_m \left|A_{\ell m}^{(2)}(k_0\hat{z}, k_0\hat{q}_\mu)\right|^2 \ \geq 0, \tag{42}$$

$$R_\ell = \frac{1}{8\pi i^\ell}\int_{-1}^{1}\mathrm{d}\mu \int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\, \mathcal{K}_3(k_0\hat{z}, k_0\hat{q}_\mu, -k_0\hat{q}_\mu; \hat{n}). \tag{43}$$

Then, exactly through fourth order,

$$\boxed{C_\ell = 4\pi v\, T_\ell^2 + v^2\big[Q_\ell + 4\pi T_\ell \,\mathrm{Re}\, R_\ell\big] + O(v^3).} \tag{44}$$

This is a reorganisation, not an evaluation, but it exposes the physics directly:

- $Q_\ell \geq 0$: the quadratic response always generates power;
- R_ℓ: the cubic response interferes with the linear response;
- at a linear node, $C_\ell = v^2 Q_\ell + O(v^3)$ (Prop. 15);
- a negative fourth-order correction away from nodes is not inconsistent.

The decompositions of Thm. 12 and Prop. 13 explain how Q_ℓ and R_ℓ are built.

What the final integrals contain. For auditability, here is every ingredient of (44), in one place:

1. *Linear transfer.* T_ℓ: Prop. 11, closed form.

2. *Second-order map kernel.* $\mathcal{K}_2 = A^{(2)}$ in harmonic space. It is the map of §6, centred as in (18) and composed with the clock.
   - Its fields are the first-order closed forms and, on the shell, the two amplitudes F_S, F_T of (16) together with ν_2, in closed form (§5).
   - Its time integrals are the ray integral over $[\bar{t}_\star, \bar{t}_o]$, which contains the nested displacement integral over $[t, \bar{t}_o]$, and the emission-endpoint and clock terms.
3. *Cubic map kernel.* $\mathcal{K}_3(k, q, -q)$.
   - The new third-order field enters linearly through $\sigma^{(3)} \ni \frac{1}{2}A_3'$. Only its $\hat{q}$-average is needed, and by Prop. 14 that average is a single scalar function. It is obtained from the lapse and Hamiltonian Green formulas with the $\hat{q}$-averaged cubic sources.
   - Everything else is a product of lower-order fields: $\sigma^{(3)}$ products, X_2, e_2, the emission shift d_2, the clock D_2 and the normalisation. These contain the second-order tensor and ν_2, which are kept in full.
   - The pair $(q, -q)$ uses the exact $K = 0$ kernel.
4. *Sky integrals.* Every term has the form $c\,\Pi(\hat{n})e^{i\hat{n}\cdot V}$, and the sky integrals are evaluated in closed form below.
5. *Remaining integrals.* What is left is the integral over $\mu \in [-1, 1]$ and the nested times. The point $\mu = -1$ is a null set (Thm. 17).

Proposition 14 (The contracted cubic response is scalar)**.** *For an isotropic spectrum, define the angle-averaged cubic metric response*

$$\bar{A}_{3,ij}(k, q; t) = \frac{1}{4\pi}\int \mathrm{d}\Omega_{\hat{q}}\, A_{3,ij}(k, q, -q; t).$$

Then

$$\bar{A}_{3,ij} = F_3(k, q; t)\Big(\hat{k}_i\hat{k}_j - \tfrac{1}{3}\delta_{ij}\Big),$$

so the vector and TT parts of the angle-averaged cubic metric response vanish.

Proof. The third-order operator depends on the total momentum k only, so averaging over $\hat{q}$ commutes with solving. After the average, $\hat{k}$ is the only preferred direction. A symmetric trace-free tensor built from $\hat{k}$ and δ alone is proportional to $\hat{k}\hat{k} - \delta/3$. □

The new cubic field work for $C^{(13)}$ is therefore one scalar problem with averaged sources. Two caveats: the lower-order tensor contributions inside the cubic sources and propagation products must be kept, and the uncontracted map $\Theta^{(3)}$ still needs all three sectors. The exploratory evaluation in App. A confirms numerically that keeping only the scalar part reproduces the full result.

Proposition 15 (Complete fourth order at linear-transfer nodes)**.** *On the shell, $C_\ell^{(13)} \propto T_\ell(k_0)$, and the cubic kernel is finite at finite times. Hence, at every nontrivial zero of $T_\ell(k_0)$, $C_\ell^{(11)} = C_\ell^{(13)} = 0$ and*

$$\boxed{C_\ell = \frac{\epsilon^4}{4}C_\ell^{(22)} + O(\epsilon^6) \qquad \text{when } T_\ell(k_0) = 0.}$$

At these points the complete fourth-order spectrum is determined by second-order quantities alone, namely the closed-form kernels of §5 and the map of §6. In terms of Thm. 12, $\frac{1}{3}\operatorname{Re} C_\ell[\theta, L_3]$, the term $\operatorname{Re} C_\ell[\theta, \theta L_2]$ and the $-C_\ell^2/2\pi$ part of $\mathcal{N}_\ell$ all vanish there. What remains is $\frac{1}{4}C_\ell[L_2, L_2] + \frac{1}{2}\operatorname{Re} C_\ell[L_2, \theta^2] + \frac{1}{4}C_\ell[\theta^2, \theta^2]$.

Near such a node a large ratio $C_\ell^{(22)}/C_\ell^{(11)}$ is expected: it is nonlinear power filling a linear spectral node. The condition $\epsilon^4 C_\ell^{(22)} \ll \epsilon^2 C_\ell^{(11)}$ is a condition for linear dominance in that multipole, not for the validity of the perturbative expansion. Exploratory values at the first nodes of T_2 and T_3 are listed in App. A.

All sky integrals in closed form. Every term of $\mathcal{K}_r$ has the form $c(\lambda,\dots)\,\Pi(\hat n)\,e^{i\hat n\cdot V}$. Here $V=\sum_a k_a\lambda_a$ collects the line-of-sight and nested times, and Π is a polynomial. By Funk–Hecke and $n_i \to -i\partial_{V_i}$:

$$\int \mathrm{d}\Omega\, P_\ell(\hat z\cdot\hat n)\,\Pi(\hat n)e^{i\hat n\cdot V} = \Pi(-i\partial_V)\big[4\pi i^\ell j_\ell(|V|)P_\ell(\hat z\cdot\hat V)\big],$$

$$\iint P_\ell(\hat n\cdot\hat n')\Pi(\hat n)e^{i\hat n\cdot V}\Pi'(\hat n')e^{-i\hat n'\cdot V'} = \Pi(-i\partial_V)\Pi'(i\partial_{V'})\big[(4\pi)^2 j_\ell(|V|)j_\ell(|V'|)P_\ell(\hat V\cdot\hat V')\big].$$

The sky *integrals* at $\ell=0$ give $4\pi j_0$; the sky *means* in the normalisation terms include the factor $1/(4\pi)$ and therefore give j_0. The remaining integrals are therefore:

- for $C^{(22)}$: μ, plus per copy one line-of-sight time, one ray time and one field-kernel time;
- for $C^{(13)}$: μ and up to five nested times.

All integrands are elementary or closed-form.

Shell locality and the real-space form. On the shell, g obeys $\nabla^2 g = -k_0^2 g$ exactly, and $\langle g(X)g(Y)\rangle = \sigma^2 j_0(k_0|X-Y|)$. Every first-order field is therefore a fixed time profile multiplying a *local* derivative of g:

$$\nu_1 = -u_1(x_0)\,g, \qquad H_{1,ij} = -Q(t)\Big(\partial_i\partial_j + \tfrac{1}{3}\delta_{ij}k_0^2\Big)g, \qquad x_0 = k_0 t/\sqrt{3},$$

with Q the first-order kernel of §6 at $k=k_0$. For a general spectrum the time profiles depend on $|k|$, and this locality is lost.

Proposition 16 (Shell locality)**.** *On the monochromatic shell:*

1. *The first-order map is a local line integral of the primordial field:*

$$\theta(\hat n) = \mathsf{P}_0\Big[\tfrac{1}{2}\int_{\bar t_\star}^{\bar t_o} Q'(t)\,\mathcal{D}_{\hat n}\,g\big(\hat n(\bar t_o - t)\big)\,\mathrm{d}t\Big], \qquad \mathcal{D}_{\hat n} = (\hat n\cdot\partial)^2 + \tfrac{1}{3}k_0^2.$$

Its correlation function is a double line-of-sight integral of an elementary function,

$$\xi(\hat n\cdot\hat n') = \frac{\sigma^2}{4}\iint \mathrm{d}t\,\mathrm{d}t'\,Q'(t)Q'(t')\,\mathcal{D}^X_{\hat n}\mathcal{D}^Y_{\hat n'}\,j_0(k_0|X-Y|)\Big|_{X=\hat n\lambda,\,Y=\hat n'\lambda'} - (\ell=0\ \textit{part}), \quad (45)$$

with $\lambda = \bar t_o - t$ and $\lambda' = \bar t_o - t'$. Its Legendre coefficients are $4\pi\sigma^2 T_\ell^2$. Consequently the real-space lognormal term $\frac{1}{2}\xi^2 - \frac{1}{2\pi}\xi\star\xi$ of (35) has elementary integrands. Its multipoles still require the viewing-angle projection (36).

2. *At second order, the* only *non-local input to the log map is the order-two trace-free metric along the background ray:*

$$L_2 = L_2^{\mathrm{loc}} - \mathsf{P}_0\Big[\tfrac{1}{2}\int_{\bar t_\star}^{\bar t_o} A'_{2,nn}\big(t,\hat n(\bar t_o - t)\big)\,\mathrm{d}t\Big].$$

Here L_2^{loc} is given by the formulas of §6 with $A_2 \to 0$. It is a local quadratic form in derivatives of g along the ray and at the observer, with fixed time profiles.

3. *At third order, the non-local inputs are A_2, A_3 and ν_2. They enter through σ_2, X_2, e_2, the emission shift d_2 and the clock shift D_2.*

Proof. (1) For a single mode, $\frac{1}{2}Q'\mathcal{D}_{\hat{n}}e^{ik\cdot X} = -\frac{1}{2}Q'k^2(\mu^2 - \frac{1}{3})e^{ik\cdot X} = -\sigma_{1,nn}$, so $\theta = -\mathsf{P}_0 I_1$, as in §6. Then (45) follows from the shell correlation of g. (2) Since $\gamma = e^{2\zeta}e^H$, the second-order shear is $\sigma^{(2)} = \frac{1}{2}A_2' + \frac{1}{2}(H_1^2)' + 2gH_1'$. Every other ingredient of I_2 and of the second-order clock composition is first order: X_1, e_1, d_1, D_1, and the frame. The lapse ν_2 enters only through d_2 and D_2, which first appear in $\Theta^{(3)}$. (3) Inspect F_3, the ray source of X_2, I_3 and D_2 in §6. □

By Wick's theorem, every contraction between local quantities is a derivative of $\sigma^2 j_0(k_0|X-Y|)$, evaluated at points on the two rays or at the observer. Therefore all of $\mathcal{N}_\ell$, and every *local–local* part of $C_\ell^{L,4}$ and $\mathcal{M}_\ell$, have correlation functions that are line-of-sight integrals of elementary functions. These parts need no integral over the primordial relative-momentum angle μ and no Green-function times. Two different angular integrations must be kept apart, however. The *viewing-angle* Legendre projection $C_\ell = 2\pi\int_{-1}^{1} P_\ell(\hat{n}\cdot\hat{n}')\, w(\hat{n}\cdot\hat{n}')\,\mathrm{d}(\hat{n}\cdot\hat{n}')$ remains, and it is not evaluated here. The relative-momentum integral over μ (equivalently the total momentum $K \in [0, 2k_0]$), and the nested Green-function times, are confined to the terms that contain A_2, A_3 or ν_2: the resonant second- and third-order field response. This is a reorganisation of the same exact integrals; the local line-of-sight integrals are not evaluated.

Theorem 17 (Existence)**.** *For all finite $0 < x_\star < x_o$, the integrals for $C_\ell^{(22)}$ and $C_\ell^{(13)}$ converge absolutely. The restriction to the shell is well defined.*

Proof. The momentum, angle and time domains are compact. The integrands are finite sums of products of phases, first-order closed forms, and the higher-order kernels and their first time derivatives. It therefore suffices to show that each kernel and its time derivative is bounded, and continuous outside a null set. The entire-function representation removes every frequency-resonance denominator, but not the inverse powers of the total momentum K introduced by the projectors. These are handled sector by sector.

1. *Lapse.* The Green function $-\frac{1}{3c}[u_2(ct)u_1(cs) - u_1(ct)u_2(cs)]$ and the homogeneous term $\frac{9c_r}{K^2}u_1(ct)$ are even entire functions of $c = K/\sqrt{3}$ (the latter equals $c_r t^2$ times one). Hence ν_r and ν_r' are continuous in K including $K = 0$, where they reduce to the $K = 0$ formulas.

2. *Scalar.* The Hamiltonian form of h_S carries K^{-2}. Use instead the equivalent longitudinal momentum-constraint form $h_S(t) = \int_0^t \left[6(\nu_r' - \nu_r/s) + 3i\, K\cdot c_M(s)/K^2\right]\mathrm{d}s$. The two agree for $K \neq 0$ by Thm. 5. Since c_M is analytic in the momenta and vanishes at $K = 0$, $|K\cdot c_M|/K^2 = O(1)$, so h_S and h_S' are bounded as $K \to 0$.

3. *Vector.* $P_\perp c_M = O(K)$, so $w = \frac{2i}{K^2}\int P_\perp c_M = O(K^{-1})$, and the metric component $K \otimes w$ is $O(1)$. On the shell $w \equiv 0$ (Prop. 7).

4. *Tensor.* The TT projector is bounded, and the Green function $\sin K(t-s)/K$ is entire and even in K.

The projectors themselves depend on $\hat{K}$, so the recombined Cartesian kernel may depend on the direction of approach at $K = 0$. This happens only at $\mu = -1$ on the shell, a null set, and boundedness is all that absolute convergence needs. Continuity of the full Cartesian kernel at $K = 0$ is not claimed, and nothing uses it. Where an exact zero total momentum occurs inside the cubic sources, namely the pair $(q, -q)$, the separately constructed exact $K = 0$ kernel is used, not a limit. The ray and line-of-sight integrals run over finite intervals, and the emission and observation times are positive. □

The theorem and the spectrum formulas therefore apply to finite times and to the monochromatic (or any compactly supported) spectrum. Arbitrary spectra without compact support are not covered: the ultraviolet behaviour is open (§9).

9 Limits

9.1 Superhorizon limit, exact in ζ

Theorem 18. *Let all momenta scale as $\kappa \to 0$ at fixed times. Then, to all orders in ϵ,*

$$\Theta(\hat{n}) = -\tau^2\, e^{-2\zeta}\ \mathrm{TF}\left[\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta\right]_o \hat{n}^i\hat{n}^j + O(\kappa^3), \qquad \tau^2 = \tfrac{1}{6}(\bar{t}_o^2 - \bar{t}_\star^2).$$

Proof. At zeroth gradient order the local seed is constant, $\zeta = \zeta_o$, and the ray direction is $e = -e^{-\zeta_o}\hat{n}$, not $-\hat{n}$. Spatial variation of ζ changes this direction by $O(\kappa)$, while H and σ are $O(\kappa^2)$. Their spatial variation along the finite ray adds one further gradient. Since $e^{-2\zeta_o}\sigma = \frac{1}{2}(e^H)'$ locally, the constant conformal factor cancels against the two direction vectors, giving

$$I_{\mathrm{lead}} = \frac{1}{2} n^i n^j [H_{ij}(\bar{t}_o, 0) - H_{ij}(\bar{t}_\star, 0)].$$

Endpoint and clock shifts are $O(\kappa^2)$ and multiply $O(\kappa^2)$ quantities. The Fermi–Walker correction also enters beyond the displayed order. Substitution of (5) gives the claimed I_{lead}. It is trace-free, so $\langle I_{\mathrm{lead}}\rangle_\Omega = 0$ and $\Theta = -I_{\mathrm{lead}} + O(\kappa^3)$.

At each fixed amplitude order the remainder is uniform in the angles on compact momentum and time domains, so the gradient expansion may be taken inside the compact integrals. This statement does not exchange an infinite amplitude series with the full nonlinear solution or its ensemble average.

The independent generator confirms the formula at first and second order, at (k, q) with general momenta. The corresponding third-order run at $(k, q, -q)$ was *not completed* (§10). At third order the formula rests on two things: the proof above, which uses only the gradient-expansion data (5) (exact in ζ and verified for arbitrary ζ), and the generator's third-order Ward identity W3. □

Geometric form. Let $f = e^{-\zeta}$ and $\gamma^{(0)} = e^{2\zeta}\delta$ be the primordial seed metric. Since $f\,\partial_i\partial_j f = -e^{-2\zeta}(\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta)$, the theorem has the equivalent forms

$$\boxed{\Theta_{\mathrm{lead}}(\hat{n}) = \tau^2 f(\partial_i\partial_j f)_{\mathrm{TF}} n^i n^j = \tau^2 R_{\langle ab\rangle}[\gamma^{(0)}] n^a n^b.} \tag{46}$$

Here $R_{\langle ab\rangle}$ is the trace-free spatial Ricci tensor in the orthonormal frame $e^{-\zeta}\partial_a$ of the *seed* metric. It is not the Ricci tensor of the physical metric at the observation event. The leading observable is a local curvature quadrupole.

Proposition 19 (Gaussian average of the leading-gradient map)**.** *Let $m_j = \int k^j \mathcal{P}(k)\,\mathrm{d}\ln k$ for $j = 0, 2, 4$, with these moments finite. For $\zeta = \epsilon g$, the exact Gaussian average of the leading-gradient expression is*

$$\boxed{C_{2,\mathrm{lead}} = \frac{16\pi}{225}\tau^4\epsilon^2 e^{8\epsilon^2 m_0}\left(m_4 + \frac{5}{3}\epsilon^2 m_2^2\right).} \tag{47}$$

For a general spectrum this is the leading-gradient coefficient when all momenta scale to zero at fixed times, not a formula for finite-wavelength modes.

Proof. Put $A_{ij} = (\partial_i\partial_j g)_{\rm TF}$ and $B_{ij} = (\partial_i g\,\partial_j g)_{\rm TF}$. The leading quadrupole tensor is $Q = -\tau^2 e^{-2\epsilon g}(\epsilon A - \epsilon^2 B)$. Isotropy and Gaussianity imply that g, A and ∇g are mutually independent: the g–Hessian covariance is pure trace and all odd-derivative covariances vanish. Thus g is independent of B and

$$\langle A:A\rangle = \frac{2}{3}m_4, \qquad \langle B:B\rangle = \frac{10}{9}m_2^2, \qquad \langle A:B\rangle = 0, \qquad \langle e^{-4\epsilon g}\rangle = e^{8\epsilon^2 m_0}.$$

The ensemble mean of Q vanishes. For a trace-free tensor Q, $\sum_m |a_{2m}|^2 = \frac{8\pi}{15}Q:Q$, so $C_2 = \frac{8\pi}{75}\langle Q:Q\rangle$, proving (47). □

On the monochromatic shell, $m_0 = \sigma^2$, $m_2 = k_0^2\sigma^2$ and $m_4 = k_0^4\sigma^2$. With $\Xi = x_o^2 - x_\star^2 = k_0^2(\bar t_o^2 - \bar t_\star^2)/3$,

$$\boxed{C_{2,\rm lead} = \frac{4\pi}{225}\epsilon^2\sigma^2\Xi^2 e^{8\epsilon^2\sigma^2}\left(1 + \frac{5}{3}\epsilon^2\sigma^2\right).} \tag{48}$$

This resums the Gaussian average of the leading-gradient map only. It is not a convergence claim for the full amplitude expansion and supplies no arbitrary-$k_0 t_o$ spectrum. Its expansion, $e^{8v}(1+5v/3) = 1 + 29v/3 + O(v^2)$ with $v = \epsilon^2\sigma^2$, reproduces the fourth-order answer. Separately, exact Wick moments give the leading $O(k_0^4)$ coefficients:

$$C_2^{(11)} = \frac{4\pi}{225}\sigma^2\Xi^2, \qquad C_2^{(22)} = \frac{272\pi}{675}\sigma^4\Xi^2, \qquad C_2^{(13)} = \frac{16\pi}{75}\sigma^4\Xi^2,$$

$$C_2 = \frac{4\pi}{225}\epsilon^2\sigma^2\Xi^2\Big[1 + \frac{29}{3}\epsilon^2\sigma^2\Big] + O(\epsilon^6) + O(k_0^6), \qquad C_{\ell\geq 3} = O(k_0^6) \text{ (the octupole is given in §9.2).}$$

Two checks: $C_2^{(11)}$ agrees with the line-of-sight transfer function $T_2 \to -\frac{2}{15}k_0^2\tau^2$, and the direct ϵ-expansion of $\langle\sum_m |a_{2m}|^2\rangle$ reproduces $\frac{1}{4}C^{(22)} + \frac{1}{3}C^{(13)}$. The fourth-order correction is a finite, positive renormalisation of the quadrupole by the factor $1 + \frac{29}{3}\epsilon^2\sigma^2$. It is independent of T_I at this order.

Dilation factorisation: what the 29/3 is made of. The map (46) is exactly covariant under the residual dilation: $\Theta_{\rm lead}[\zeta + c] = e^{-2c}\,\Theta_{\rm lead}[\zeta]$, and it carries two spatial derivatives. A constant shift of ζ is therefore *exactly* the momentum dilation $k \to e^{-c}k$. In the Gaussian average, the observer's own value $\zeta_o = \epsilon g(0)$ is independent of the trace-free Hessian and the gradient at the observer, so it acts as a dilation of everything else:

$$\boxed{\begin{aligned} C_{2,\rm lead} &= \underbrace{\langle e^{-4\zeta_o}\rangle}_{\text{observer dilation}} \frac{16\pi}{225}\tau^4\epsilon^2\Big(\underbrace{m_4}_{\text{Hessian}} + \underbrace{\tfrac{5}{3}\epsilon^2 m_2^2}_{\text{gradient}^2}\Big), \\ \langle e^{-4\zeta_o}\rangle &= \langle e^{-\zeta_o D}\rangle\big|_{D=4} = e^{\frac{1}{2}\epsilon^2 m_0 D^2}\big|_{D=4} = e^{8\epsilon^2 m_0}. \end{aligned}} \tag{49}$$

Here $D = k\partial_k$ acts on the degree-four momentum norms. Three consequences follow.

1. *Origin of the coefficients.* The fourth-order coefficient splits as $\frac{29}{3} = 8 + \frac{5}{3}$. The $8 = \frac{1}{2}D^2|_{D=4}$ is the observer dilation; it is shared equally between $\frac{1}{4}C^{(22)}$ (4) and $\frac{1}{3}C^{(13)}$ (4). The $\frac{5}{3}$ is pure $\frac{1}{4}C^{(22)}$. It is exactly the Gaussian average of the gradient-squared quadrupole (22), the effect isolated by [1].

2. *Same operator as in the infrared.* The $O(\epsilon^2)$ term of $\langle e^{-\zeta_o D}\rangle$ is $\frac{1}{2}\epsilon^2 m_0 D^2$. This is the operator of Thm. 22, $\frac{1}{2}\sigma_L^2 D^2$ acting on the short-mode spectrum, which is equivalent to acting on the response after integration by parts in $\ln k$. In the superhorizon limit every mode is "long" relative to the horizon, and the field at the observer dilates its own gradients.

3. *What is, and is not, a convention.* A genuine constant relabelling $X' = e^c X$, $\zeta'(X') = \zeta(e^{-c}X') - c$ changes the mean, $\mu' = \mu - c$, and rescales the moments, $m'_j = e^{-jc} m_j$, but it leaves the variance $m'_0 = m_0$ unchanged. The full spectrum is invariant: the deterministic factor $e^{-4\mu}$ compensates the rescaled moments. The stochastic factor $e^{8\epsilon^2 m_0}$ is the average of the observer's *random* local dilation, and it survives every such relabelling. For the ensemble specified in the problem (Gaussian ζ with the chart-shell k_0 and σ^2 held fixed), the fourth-order coefficient is therefore $\frac{29}{3}$, observer dilation included. The ratio

$$\frac{C_{2,\mathrm{lead}}}{\frac{16\pi}{225}\tau^4\epsilon^2\langle e^{-4\zeta_o}\rangle m_4} = 1 + \frac{5}{3}\epsilon^2\frac{m_2^2}{m_4} \xrightarrow{\text{shell}} 1 + \frac{5}{3}\epsilon^2\sigma^2 \tag{50}$$

   is a useful *dilation-factored* ratio, valid to all orders in ϵ at leading gradient order. It is not a replacement answer.

Equivalently, write $k_{\mathrm{phys}} = e^{-\zeta_o}k$ for the wavenumber in the seed's orthonormal frame at the observer, and put $\Xi_{\mathrm{phys}} = e^{-2\zeta_o}\Xi$. Then, as an identity on the shell,

$$C_{2,\mathrm{lead}} = \frac{4\pi}{225}\epsilon^2\sigma^2\,\langle\Xi^2_{\mathrm{phys}}\rangle\Big(1 + \frac{5}{3}\epsilon^2\sigma^2\Big). \tag{51}$$

Holding a locally inferred scale fixed instead of the chart-shell k_0 would be a *different*, conditional specification of the ensemble. It is not the one posed. `solution/check_dilation_factorisation.py` verifies the covariance exactly in ζ, the mean-shift/dilation equivalence and the Gaussian dilation operator.

9.2 Leading octupole: a curvature-gradient multipole

For every $\ell \geq 2$, the lowest powers of $s'(x) = x/3 + O(x^3)$ and $w_\ell(y) = \ell(\ell-1)y^{\ell-2}/(2\ell+1)!! + O(y^\ell)$ in (27) give

$$T_\ell(k) = -\frac{k^\ell(\bar{t}_o - \bar{t}_\star)^{\ell-1}\big[\bar{t}_o + (\ell-1)\bar{t}_\star\big]}{3\,(2\ell+1)!!} + O(k^{\ell+2}). \tag{52}$$

For $\ell = 2$ this is $T_2 \to -\frac{2}{15}k^2\tau^2$. For $\ell = 3$ it is $T_3 \to -\frac{2}{35}k^3\beta_3$, with

$$\beta_3 = \frac{1}{3}\int_{\bar{t}_\star}^{\bar{t}_o} t\,(\bar{t}_o - t)\,\mathrm{d}t = \frac{(\bar{t}_o - \bar{t}_\star)^2(\bar{t}_o + 2\bar{t}_\star)}{18}.$$

Theorem 20 (Leading octupole, exact in ζ)**.** *Let all momenta scale as $\kappa \to 0$ at fixed times. To every order in ϵ, the $\ell = 3$ part of the map is*

$$\boxed{\Theta_{3,\mathrm{lead}}(\hat{n}) = \beta_3\,\big[\nabla_{\langle a}R_{bc\rangle}\big]_o\,n^a n^b n^c + O(\kappa^4), \qquad \nabla_{\langle a}R_{bc\rangle} = \frac{f}{2}\big[\partial_a\partial_b\partial_c(f^2)\big]_{\mathrm{STF}},} \tag{53}$$

where $f = e^{-\zeta}$. The curvature, the covariant derivative and the orthonormal components are those of the seed metric $\gamma^{(0)} = e^{2\zeta}\delta$, and $\langle\cdots\rangle$ denotes the symmetric trace-free part.

Proof. 1. *Coordinates.* Use seed Riemann normal coordinates at the observer. This is a time-independent relabelling of the comoving labels, so it is allowed at fixed physical labels. The Christoffel symbols vanish at the observer and are $O(\kappa^2)$ along the ray. Ray deflection and displacement therefore enter only at $O(\kappa^4)$, and so do the endpoint, the clock, the frame rotation and the nonlinear normalisation terms.

2. *Shear.* The leading shear is covariant, $\sigma_{ab} = -\frac{t}{3} R_{\mathrm{TF},ab}$, by (5).

3. *Expansion along the ray.* Expanding along the radial ray,

$$\Theta = \frac{1}{3} \int t \left[R_{\mathrm{TF}} + (\bar{t}_o - t)\, n^c \nabla_c R_{\mathrm{TF}} \right]_{nn} \mathrm{d}t + O(\kappa^4).$$

The first term is the quadrupole (46). The second term is $\beta_3 \nabla_{\langle a} R_{bc\rangle} n^a n^b n^c$ plus a dipole, which by the contracted Bianchi identity is proportional to $n \cdot \nabla R$. The dipole is removed by the final projection and does not affect $\ell = 3$.

4. *Conformal identity.* The conformal identity in (53) is proved for generic ζ in `solution/check_gradient_octupole.py`.

□

With the moments m_j of Prop. 19 (now including m_6), the same independence argument gives an exact Gaussian average. Write $\Theta_{3,\mathrm{lead}} = \beta_3 Q_{abc} n^a n^b n^c$ with $Q_{abc} = e^{-3\epsilon g}\big[-\epsilon g_{abc} + 6\epsilon^2 g_{(ab} g_{c)} - 4\epsilon^3 g_a g_b g_c \big]_{\mathrm{STF}}$, by (53). The three coefficients come from three facts:

- $\langle |g_{\langle abc\rangle}|^2 \rangle = \frac{2}{5} m_6$;
- the $\ell = 3$ part of the trace-free Hessian $\otimes$ gradient carries $\frac{7}{15}$ of $\langle A{:}A \rangle \langle |\nabla g|^2 \rangle$;
- $\langle |(\nabla g)^{\otimes 3}_{\mathrm{STF}}|^2 \rangle = \frac{2}{5} \langle |\nabla g|^6 \rangle = \frac{14}{9} m_2^3$.

With $\sum_m |a_{3m}|^2 = \frac{8\pi}{35} Q{:}Q$, the result is

$$\boxed{C_{3,\mathrm{lead}} = \frac{16\pi}{1225} \beta_3^2 \epsilon^2 e^{18\epsilon^2 m_0} \left[m_6 + 28\epsilon^2 m_2 m_4 + \frac{560}{9} \epsilon^4 m_2^3 \right].} \tag{54}$$

On the shell, with $v = \epsilon^2 \sigma^2$,

$$\boxed{C_{3,\mathrm{lead}} = \frac{4\pi}{3675} (x_o - x_\star)^4 (x_o + 2x_\star)^2 \epsilon^2 \sigma^2 e^{18v} \Big(1 + 28v + \frac{560}{9} v^2 \Big),} \tag{55}$$

and, through the requested order,

$$C_3 = \epsilon^2 C_3^{(11)} \big[1 + 46\epsilon^2 \sigma^2 \big] + O(\epsilon^6) + O(k_0^8), \qquad \frac{C_3^{(22)}}{C_3^{(11)}} = 148\sigma^2, \quad \frac{C_3^{(13)}}{C_3^{(11)}} = 27\sigma^2.$$

The linear term agrees with $4\pi\sigma^2 T_3^2$ from (52). The structure parallels the quadrupole:

- the observer dilation is $e^{\frac{1}{2}\epsilon^2 m_0 D^2}$ at degree $D = 6$, giving e^{18v} (compared with $D = 4$ and e^{8v} for the quadrupole);
- $46 = 18 + 28$, the second term being the Hessian–gradient interference;
- the v^2 term is the cubed gradient.

Bounds for general spectra. For any nonnegative spectrum with finite, nonzero moments, the leading-gradient fourth-order factors are

$$\frac{C_{2,\mathrm{lead}}}{\epsilon^2 C^{(11)}_{2,\mathrm{lead}}} = 1+\epsilon^2 b_2, \quad b_2 = m_0\Big(8+\tfrac{5}{3}r_2\Big), \qquad \frac{C_{3,\mathrm{lead}}}{\epsilon^2 C^{(11)}_{3,\mathrm{lead}}} = 1+\epsilon^2 b_3, \quad b_3 = m_0(18+28r_3),$$

up to $O(\epsilon^4)$, with shape parameters $r_2 = m_2^2/(m_0 m_4)$ and $r_3 = m_2 m_4/(m_0 m_6)$. The Cauchy–Schwarz inequalities $m_2^2 \le m_0 m_4$ and $m_4^2 \le m_2 m_6$ give $0 \le r_3 \le r_2 \le 1$. Hence

$$\boxed{8\,m_0 \le b_2 \le \tfrac{29}{3}\,m_0, \qquad 18\,m_0 \le b_3 \le 46\,m_0,} \tag{56}$$

with equality at the upper ends exactly for the monochromatic shell. At fixed primordial variance, the shell therefore maximises both relative leading-gradient corrections. The lower ends are approached by very broad spectra, for which only the observer dilation survives.

This evaluates the leading $O(k_0^6)$ octupole only, not the finite-wavelength spectrum, and the exponential resums the leading-gradient expression only. The cross-check by the independent ray generator (`observable_generator.py`) was not run at this order.

9.3 Late time, first order

Proposition 21. *For $\ell \ge 2$ and fixed $x_\star$, $T_\ell(x_\star, x_o) = O(\ln x_o/x_o)$ as $x_o \to \infty$. Hence $C_\ell^{(11)} \to 0$.*

Proof. Write $T_\ell = -3\int_0^{x_o - x_\star} s'(x_o-u) w_\ell(\sqrt{3}u)\mathrm{d}u$, with $w_\ell = j_\ell/3 + j_\ell''$. Two integrations by parts give $\int_0^\infty e^{iu} w_\ell(\sqrt{3}u)\mathrm{d}u = 0$ exactly for $\ell \ge 2$, because $j_\ell(0) = j_\ell'(0) = 0$. This is the observer-local sound-cone cancellation. Since $s'(x) = \sin x + O(1/x)$, the $O(1)$ oscillation therefore vanishes. The remaining pieces are oscillatory with distinct frequencies $\sqrt{3}\pm 1$ and amplitudes $O(1/x_o)$, giving $O(\ln x_o/x_o)$ by integration by parts. □

At fourth order the late-time limit is non-uniform at the resonances $\mu = \pm 1$ and $\mu = -1/3$. A pointwise late-time limit therefore cannot simply be interchanged with the angular integral without a uniform estimate or an explicit treatment of the resonant regions. Matched asymptotics in angle and time, or an analytically subtracted representation, would be legitimate routes. The evaluation is [open].

9.4 General spectra

Theorem 22 (Infrared)**.** *For spectra smooth on $(0,\infty)$, consider the $q \to 0$ region. By the Ward identities, $K_2(k,q) = -k\cdot\partial_k K_1(k) + O(q)$ and $K_3(k,q,-q) = (k\cdot\partial_k)^2 K_1(k) + O(q^2)$. Here the kernels are analytic in q at fixed finite times, and the absence of an $O(q)$ term in K_3 follows from its symmetry under $q \to -q$. The $O(q)$ cross term in $|K_2|^2$ is odd in $\hat{q}$ and integrates to zero.*

Both the q-long and the k-long regions of $C^{(22)}$ contribute. Let $\sigma_L^2 = \int_{q<\Lambda} P$ and $D = k\cdot\partial_k$. The infrared-sensitive part of $\frac{1}{4}C^{(22)} + \frac{1}{3}\operatorname{Re} C^{(13)}$ is then

$$\begin{aligned}&\frac{\sigma_L^2}{2\ell+1}\int_k P(k)\Big[\sum_m |DA^{(1)}_{\ell m}|^2 + \operatorname{Re}\sum_m A^{(1)}_{\ell m} D^2 A^{(1)*}_{\ell m}\Big] \\ &\qquad = \frac{\sigma_L^2}{2(2\ell+1)}\int_k \Big[\sum_m |A^{(1)}_{\ell m}|^2\Big](D+3)^2 P(k).\end{aligned}$$

The equality uses two integrations by parts in $\ln k$ *and assumes vanishing radial boundary terms; for spectra with explicit cutoffs those boundary terms must be retained. Under that assumption the expression vanishes for scale-invariant* $P \propto k^{-3}$*. Otherwise it is the physical rescaling of the short-mode spectrum by the long-mode variance. The remainder is* $O(q^2)P(q)$*, which is integrable for* $\mathcal{P}(q)$ *with* $\int_0 \mathcal{P}(q)q^2 \,\mathrm{d}\ln q < \infty$*.*

Ultraviolet [power counting only]. At fixed finite times:

- the first-order kernels are bounded uniformly in k;
- each order of the sources carries at most two spatial derivatives;
- each ray displacement carries one.

Counting powers on this basis gives $|\mathcal{K}_2(k,q)| \lesssim q^2$ and $|\mathcal{K}_3(k,q,-q)| \lesssim q^4$ as $q \to \infty$. That suggests absolute convergence of $C^{(22)}$ and $C^{(13)}$ whenever $\int^\infty \mathcal{P}(q)q^4 \,\mathrm{d}\ln q < \infty$.

This is *not* a proof. A rigorous bound needs uniform estimates of the lapse Green function near $s \to 0$ (where $|G_N|$ grows like $1/(K^3 ts)$) against sources whose small-s Taylor coefficients grow with q. The true large-q behaviour for a scale-invariant spectrum, including possible cancellations, is [open].

10 Verification ledger

All checks are exact symbolic computations (SymPy 1.14). None uses floating point or quadrature as evidence. "Generic 2-jets" means that every function value and derivative is an independent symbol, so a zero residual is a proof.

The checks are of different kinds, and the ledger says which kind each one is:

- *general proofs*: generic jets, arbitrary fields, closed-form identities;
- *finite exact series tests*: all field equations through a stated order in t;
- *exact closed-form substitutions*: completed for the lapse equation at second order only.

In particular, the assembled third-order kernel at $(k,q,-q)$ is checked by an exact series through $O(t^6)$ and by Ward identities in κ. It is not checked by substitution at arbitrary finite time. The exploratory numerics of App. A add finite-k Ward tests, but they are not evidence.

Check	Script and scope
Lemmas L1–L5; generic field equations	`solution/foundations/check_exact_lemmas.py`, `check_fw_conformal.py`, `check_exact_field_equations.py`, `check_field_equations_generic.py` (generic 2-jets)
Gradient-expansion data (5)	`solution/foundations/check_initial_data_gradient.py` (exact in ζ)
Mixed trace-free form (T)	`solution/check_evolution_form.py` (generic 2-jets)
First order, two gauges, all ℓ	`solution/foundations/check_first_order.py`; `solution/first_order_engine_check.py`

Check	Script and scope
Entire-function T_ℓ (Prop. 11)	`solution/transfer_entire.py 2 3 4 5 6`: the $x_\star$-derivative of (29) equals the integrand exactly, as an identity of exponential polynomials, and the representation vanishes at $x_\star = x_o$. A printed high-precision comparison is supplementary only.
Green functions, $K \to 0$	`solution/check_green.py`
2nd-order sources and operators	`solution/second_order.py`
2nd order, all 10 equations	`solution/second_order_consistency.py`: exact Taylor series through $O(t^8)$, general K and $K = 0$
2nd-order closed forms	`solution/closed_second_order.py`: every antiderivative by exact differentiation; (N) by closed-form substitution (`--check N`); (M) and (T) via the theorem and the series test
3rd-order sources and operators	`solution/third_order.py`
3rd order, all 10 equations	`solution/fast_series.py 8`: the pair kernels, including the exact $K = 0$ pair, then the triple at $(k, q, -q)$ with general momenta. The *full nonlinear* equations are recomputed with the solved kernels and vanish through $O(t^6)$. The arithmetic is exact (sparse polynomials over $\mathbb{Q}(i)$ with formal inverse momenta; denominators are cleared exactly at the zero test). Negative controls (a wrong c_3, a corrupted $K = 0$ pair kernel) fail at the expected orders. The run takes about 70 s. `third_order_consistency.py` is the slower SymPy reference implementation of the same test.
Dilation Ward identities (fields)	`solution/check_ward_dilation.py`, `check_ward_dilation3.py` (closed form; negative controls)
Observable expansion formulas	`solution/check_observable_expansion.py` (arbitrary fields, general jets)
Observable Ward identities	`solution/observable_generator.py W2|W3` through $O(\kappa^4)$
Superhorizon observable	`solution/observable_generator.py SH`: orders 1 and 2, general (k, q). *Not completed:* `SH3` (third order at $(k, q, -q)$). Its fields are built by the fast engine in about 12 s, but the generator's SymPy assembly of the observable did not finish within 50 minutes. Porting the generator to the fast arithmetic would remove this; it was not done.
Superhorizon C_2	`solution/superhorizon.py`

Check	Script and scope
Report identities	`solution/check_report_identities.py`: background residual Bianchi identities and zero-mode constraint matching; conformal driver and cubic commutator shear; Fermi–Walker rotation; centred normalisation and polarised map kernels; the two-norm spectrum; general-moment and monochromatic leading-gradient resummation. These are exact symbolic identities, not an arbitrary-time cubic-observable verification.
Gradient-mode tests	`solution/check_gradient_modes.py`: exact special-conformal Jacobian and flat seed; zero shear and hence zero map for the completed adiabatic profile; quadratic completion and its conversion to the convention of [1]; nonzero uncompensated-gradient quadrupole; plane-wave soft generator and the leading-gradient check of (23). The general finite-time Ward identity is proved by covariance, not by full-kernel substitution.
Log-map decomposition	`solution/check_logmap_decomposition.py` (about 2 min): (31) and the $3j$ lognormal term (33) for $\ell = 1, \dots, 4$, in a band-limited Gaussian sky with non-Gaussian L_2, L_3; exact sphere integrals and Gaussian moments
Shell two-amplitude form (Prop. 7)	`solution/check_shell_reduction.py`: $c_{M,z} = c_{T,xz} = c_{T,yz} = 0$ for general momenta; $K \times c_M = 0$ on the shell identically in t and μ; off-shell negative control
Shell locality	`solution/check_shell_locality.py`: the local form of the shell first-order fields and of θ; $\sigma^{(2)} = \frac{1}{2}A_2' + \frac{1}{2}(H_1^2)' + 2gH_1'$ for generic matrices. Item (3) of Prop. 16 is by inspection.
Leading octupole; $\mathcal{M}_\ell$ cancellation	`solution/check_gradient_octupole.py`: the conformal identity $\nabla_{\langle a}R_{bc\rangle} = \frac{f}{2}[\partial^3 f^2]_{\mathrm{STF}}$ for generic ζ; the leading T_ℓ coefficient (52) for $\ell = 2, \dots, 7$; β_3; exact isotropic Wick moments giving (54) and the shell ratios $148\sigma^2$, $27\sigma^2$; $\langle\theta\theta L_2\rangle = 0$ at leading gradient order. The octupole map itself rests on the proof of Thm. 20.
Dilation factorisation	`solution/check_dilation_factorisation.py`: $\Theta_{\mathrm{lead}}[\zeta + c] = e^{-2c}\Theta_{\mathrm{lead}}[\zeta]$ exactly in ζ and scaling for a generic cubic seed; mean shift $\equiv$ dilation of the moments; invariant ratio (50); Gaussian average of $e^{-\zeta_o D}$; superhorizon observer-centred kernels of Prop. 13 ($\mathcal{K}_2^c = -\tau^2\,\mathrm{TF}(kq+qk)_{nn}$, $\mathcal{K}_3^c(k,q,-q) = 0$)

Check	Script and scope
Normalisation; Wick and Fourier measures	`solution/foundations/normalization_expansion.py`; `solution/foundations/check_algebra.py`

11 What remains unevaluated

1. For general k_0, $C_\ell^{(22)}$ and $C_\ell^{(13)}$ are given as exact absolutely convergent integrals: over μ and the nested times, with closed-form integrands, and with all sky integrals evaluated. The compact source, map and two-norm representations retain these same integrals. No further general cancellation or smaller evaluated closed form has been established; this is not a proof that none exists. The integrals are *not* evaluated beyond the superhorizon limit. The expanded ancillary equations remain available as a reference rather than being embedded in the report. The literature-inspired (24) fixes the first two soft coefficients but does not determine the comparable-momentum remainder or the full cubic response. After the log-map decomposition (Thm. 12), the unevaluated content is $C_\ell^{L,4}$ and $\mathcal{M}_\ell$. At zeros of $T_\ell(k_0)$ the cubic term drops out and the complete fourth-order spectrum is $\frac{1}{4}\epsilon^4 C_\ell^{(22)}$, which needs second-order quantities only (Prop. 15). By Prop. 16, their local–local parts are line-of-sight integrals of elementary functions; these are not evaluated. Only the terms with A_2, A_3 or ν_2 carry the angular and Green-function integrals.

2. The leading superhorizon quadrupole ($O(k_0^4)$) and octupole ($O(k_0^6)$) are evaluated. The next corrections are not computed: the $O(k_0^6)$ correction to C_2 at fourth order, and the $O(k_0^8)$ correction to C_3. They need the $O(t^4)$ gradient data and the $O(\kappa^4)$ map (ray deflection, endpoint, clock and nonlinear normalisation).

3. The second-order tensor kernel has not been compared term by term with the scalar-induced tensor kernels of §7.2.

4. The late-time limit at fourth order (non-uniform at the resonances) is not evaluated.

5. Ultraviolet convergence for spectra without compact support is established by power counting only, not by a rigorous bound.

6. The infrared statement (Thm. 22) assumes that the kernels are analytic in q at $q = 0$ at fixed finite times. This holds for the integral representations by construction, but the uniformity of the expansion in k over unbounded spectra is not proved.

7. The results are established for finite emission and observation times and for the monochromatic (or any compactly supported) spectrum. Spectra without compact support need the ultraviolet and infrared estimates above.

8. The perturbative domain is treated order by order: a single emission intersection and no caustics. There is no statement about convergence in ϵ. In particular, (48) is an exact average of a leading-gradient expression, not a justification for interchanging the full nonlinear solution, an infinite amplitude series and an ensemble average.

A Exploratory numerics at finite k_0 (not evidence)

The problem forbids numerics as a substitute for derivation, and nothing in the main text relies on this appendix. It records an exploratory evaluation, made to look for structure at finite k_0 that a later analytical argument might prove, so that readers can build on it.

What is computed. By Thm. 12, every part of the fourth-order spectrum except $\frac{1}{3}\operatorname{Re} C_\ell[\theta, L_3]$ needs only second-order quantities. Those parts are $\frac{1}{4}C_\ell^{(22)}$, $\mathcal{N}_\ell$, $\mathcal{M}_\ell$ and $\frac{1}{4}C_\ell[L_2, L_2]$. `solution/numerics/explore_fourth_order.py` evaluates them for $\ell = 2$ on the shell ($k_0 = 1$, $\sigma = 1$), at fixed $\bar{t}_o/\bar{t}_\star = 3$ and $\bar{t}_I = \bar{t}_\star/2$. The ingredients are:

- the exact second-order sources of `second_order.py`, inserted into the Green formulas of §5 and integrated by quadrature;
- the ray, emission, clock and normalisation formulas of §6;
- Gauss–Legendre quadrature in μ and on the sphere.

Validation.

1. The first-order part reproduces the entire-function T_2 of Prop. 11 to 10^{-6} for $x_o \le 6$, and to 1.2×10^{-4} at $x_o = 12$ (ray grid $n_t = 160\, x_o$).
2. $\frac{1}{4}C_2[\theta^2, \theta^2]$ equals the first term of $\mathcal{N}_2$ to all printed digits.
3. The dilation Ward identity holds at finite k ($x_o = 1.5$) for the complete second-order map: $[\mathcal{K}_2(k,q) + \mathcal{K}_2(k,-q)]/2 = -k\partial_k \mathcal{K}_1$, with residual 4.5×10^{-3} at $q = 0.04$ and 1.1×10^{-3} at $q = 0.02$, i.e. $O(q^2)$. As negative controls, dropping the ray terms or A_2 gives $O(1)$ failures. The clock and endpoint terms vanish in this limit and are not tested by it.
4. The clock shift agrees with a direct root of the exact clock equation to eight digits.
5. The moving-emission endpoint term agrees, to all printed digits, with the mixed $z_a z_b$ coefficient obtained by solving the emission condition $\ln t_e + \nu(t_e, X(t_e)) = \ln \bar{t}_\star$ directly for real two-mode fields.
6. A_2' is computed analytically from the Green representation: the boundary terms of the cumulative integrals cancel, ν_2'' comes from the lapse equation, and c'_{Ham} is differentiated symbolically. It agrees with a finite-difference derivative to 10^{-4}.
7. Moving the early-time cutoff of the Green integrals from $10^{-3}\bar{t}_o$ to $2 \times 10^{-5}\bar{t}_o$ changes the results in the seventh digit.
8. The ray quadrature error scales as n_t^{-2}. On a coarse grid ($n_t = 320$) it gives a 0.4% error in T_2 at $x_o = 12$. With $n_t = 160\, x_o$, the first-order error is below 10^{-4} for $x_o \le 12$. Grid refinement in μ, the sphere and the ray changes a node value at $x_o \approx 7.9$ only in the sixth digit.
9. As $x_o \to 0$, $\frac{1}{4}C_2^{(22)}/\sigma^2 C_2^{(11)}$ approaches $\frac{17}{3}$.
10. At every node of T_ℓ below, the pipeline gives $C_\ell^{(11)} \sim 10^{-12}$ and $\operatorname{Re} C_\ell[\theta, \theta L_2] \sim 10^{-6}$. Both vanish exactly there (Prop. 15).

The convergence tests were made at $x_o \le 3$, at one node near $x_o = 7.9$, and at $x_o = 12$ on both trajectories, where $24 \to 40$ μ nodes and a $1.5\times$ finer ray grid change $\frac{1}{4}C_2^{(22)}$ only in the sixth digit. At $x_o = 16$ the μ quadrature needs 32 nodes, and the $x_o = 16$ rows use them (see the note below the tables).

Results. Absolute values with $\sigma = 1$ and $\ell = 2$, along the fixed-ratio trajectory $x_\star = x_o/3$, $x_I = x_o/6$. The first-order error is at most 1.2×10^{-4}.

x_o	$C_2^{(11)}$	$\frac{1}{4}C_2^{(22)}$	$\frac{1}{4}C_2[L_2, L_2]$	$\mathcal{M}_2$	$\mathcal{N}_2$
0.5	0.00243	0.00867	0.00867	1.9×10^{-6}	-6.0×10^{-7}
1	0.0258	0.0172	0.0177	-2.9×10^{-4}	-6.7×10^{-5}
1.5	0.0596	0.388	0.396	-0.0119	-3.1×10^{-4}
2	0.0423	2.61	2.64	-0.0453	9.5×10^{-5}
3	0.116	6.26	6.32	-0.320	-0.0012
4	0.958	8.32	8.65	-2.10	-0.048
6	0.0458	5.23	4.96	0.697	0.126
8	0.00059	6.80	4.65	1.26	0.746
12	0.00036	2.43	2.44	-0.057	0.020
16†	0.0533	1.29	1.33	0.267	0.035

Fixed-emission trajectory, $x_\star = 1$ and $x_I = \frac{1}{2}$:

x_o	$C_2^{(11)}$	$\frac{1}{4}C_2^{(22)}$	$\frac{1}{4}C_2[L_2, L_2]$	$\mathcal{M}_2$	$\mathcal{N}_2$
2	0.0340	2.12	2.14	-0.032	-6.7×10^{-5}
4	0.946	8.31	8.53	-1.98	-0.028
8	0.0014	4.21	2.57	0.907	0.563
12	0.0302	1.76	1.63	-0.304	0.041
16†	0.00091	0.787	0.719	-0.061	0.047

†At $x_o = 16$, 32 μ nodes are needed. With 24 nodes, which suffice at $x_o \le 12$, $\frac{1}{4}C_2^{(22)}$ comes out 3% low (fixed ratio: 1.2514 against 1.28688) and 15% low (fixed emission: 0.66617 against 0.787269). Increasing to 48 nodes leaves both values unchanged to six digits, and a $1.5\times$ finer ray grid has no effect. The relative-momentum integrand therefore acquires structure between $x_o = 12$ and 16 that 24 nodes do not resolve. The $x_o = 16$ rows above use 32 nodes.

Complete fourth-order spectra at linear-transfer nodes (Prop. 15). At these points $C_\ell = \epsilon^4 \sigma^4 \, c_\ell$ exactly through fourth order, with:

	trajectory	nodes x_o of T_ℓ	$c_\ell = \frac{1}{4}C_\ell^{(22)}/\sigma^4$
$\ell = 2$	$x_\star = x_o/3$	2.49867, 6.38035, 7.90609	5.686, 3.982, 7.285
$\ell = 2$	$x_\star = 1$	2.51338, 6.33088, 7.50682	5.538, 2.732, 6.222
$\ell = 3$	$x_\star = x_o/3$	3.05037, 7.00073, 8.69752	3.886, 4.334, 2.353
$\ell = 3$	$x_\star = 1$	3.05208, 6.92329, 8.48546	3.920, 2.582, 1.156

The node positions are roots of the entire-function T_ℓ. Further nodes for $x_o \le 14$ are at 9.46677 and 11.96208 ($\ell = 2$, fixed ratio), at 8.52178, 10.89081 and 13.06476 ($\ell = 2$, $x_\star = 1$), at 10.60041 and 13.04624 ($\ell = 3$, fixed ratio), and at 9.38590, 11.66731 and 13.83506 ($\ell = 3$, $x_\star = 1$).

Observations. The following are leads for analysis, not conclusions.

1. *Linear dominance, not perturbative breakdown.* Inside the horizon, $\frac{1}{4}C_2^{(22)}$ stays $O(1)\sigma^4$ while $C_2^{(11)}$ is smaller and passes through nodes of T_2. With $r_{22} = C_2^{(22)}/(4\sigma^2 C_2^{(11)})$, the quadratic contribution is smaller than the linear one only if $\epsilon^2\sigma^2\, r_{22} \ll 1$, and $r_{22} \gg 1$ for $x_o \gtrsim 2$. Away from nodes the 13 term also enters the total correction. At a node, the first nonzero prediction is the fourth-order one (Prop. 15). This is a statement about which order dominates a given multipole. It does not establish breakdown of the expansion, which would need the next order. Away from nodes, the missing 13 term could still change the fourth-order correction materially.

2. *Measurement nonlinearity.* $\mathcal{N}_2+\mathcal{M}_2$ is below about 5% of $\frac{1}{4}C_2[L_2, L_2]$ for $x_o \le 3$, and 25–45% at $x_o = 4$–8.

3. *Large cancelling pieces.* At small x_o almost all of $\frac{1}{4}C_2[L_2, L_2]$ comes from the A_2 part. At larger x_o the local part, the A_2 part and their interference are each large and strongly anticorrelated: at $x_o = 12$ (fixed ratio) they are 12.5, 17.4 and -27.5, summing to 2.4. This suggests recombining the amplitudes *before* squaring. For example, $\int A'_{2,nn} e^{iK\cdot\hat{n}(\bar{t}_o - t)}\mathrm{d}t = \left[A_{2,nn} e^{iK\cdot\hat{n}(\bar{t}_o - t)}\right]_{\bar{t}_\star}^{\bar{t}_o} + iK\cdot\hat{n}\int A_{2,nn} e^{iK\cdot\hat{n}(\bar{t}_o - t)}\mathrm{d}t$ turns the field term into endpoint pieces. Those could combine with the observer-frame, ray and emission terms, before the separately squared norms are formed.

4. *Resonances.* The μ-integrand is smooth at these finite times, as the entire-function representation guarantees. It peaks at $K \approx 0.8$–$1.0\, k_0$ in the tested configurations. Neither fact settles the asymptotics. A resonant layer of width $\propto 1/t$ is regular at every finite time yet can dominate at late times, so a matched asymptotic analysis in angle and time is needed before any late-time claim. In these runs, $\frac{1}{4}C^{(22)}$ divided by $T_2(K)^2$ is not universal in x_o.

5. *Two different limits.* The main table follows $x_\star = x_o/3$ and $x_I = x_o/6$: increasing k_0 at fixed physical labels, which is the k_0-dependence the problem asks for. The late-time limit of §9 instead keeps $x_\star$ and x_I fixed. The second table follows that trajectory with $x_\star = 1$, $x_I = \frac{1}{2}$. The two must not be conflated.

The cubic term and the complete fourth-order spectrum. `solution/numerics/explore_third_order.py` evaluates the remaining piece, $\frac{1}{3}\,\mathrm{Re}\, C_\ell[\theta, L_3]$, at $(k, q, -q)$. Its ingredients are:

- the exact third-order sources of `third_order.py`, together with the third-order Hamiltonian source, extracted for this purpose;
- the numerical second-order pair fields, with the exact $K = 0$ kernel for the pair $(q, -q)$;
- the Green formulas of §5 for A_3.

The map is not taken from the hand-expanded formulas. It is solved from its definitions by a numerical multilinear engine: the ray equation, the fields at the displaced ray, the emission root, the clock equation, the composition with the random observation time, and the normalisation. The checks are:

1. The engine's second-order map reproduces the hand-formula evaluator above to 1.2×10^{-6}, and its first-order map to 3×10^{-9}. These are two independent implementations.

2. The third-order dilation Ward identity $\mathcal{K}_3(k,q,-q) \to (k\partial_k)^2\mathcal{K}_1$ holds at finite k ($x_o = 1.5$). The residual is 3.3×10^{-2} at $q = 0.08$ and 8.2×10^{-3} at $q = 0.04$, i.e. $O(q^2)$. Dropping A_3 gives a 46% failure.

3. Keeping only the scalar part of A_3 gives the same $\frac{1}{3}\operatorname{Re} C_2[\theta, L_3]$ as keeping its vector and TT parts, to every printed digit. This confirms that the contracted cubic response is purely scalar.

4. As $x_o \to 0$, $\frac{1}{3}\operatorname{Re} C_2[\theta, L_3]/\sigma^2 C_2^{(11)}$ is 3.720 at $x_o = 0.15$ and 2.895 at 0.3; removing the $O(x_o^2)$ term extrapolates to 3.995, against the expected 4 (§9.1).

5. At the node $x_o = 2.49867$ of T_2 the cubic term is -2×10^{-5}, consistent with its exact vanishing (Prop. 15).

With $C_2^{(4)} \equiv \frac{1}{4}C_2^{(22)} + \frac{1}{3}\operatorname{Re} C_2^{(13)}$, assembled from Thm. 12 as $\frac{1}{4}C_2^{(22)} + \frac{1}{3}\operatorname{Re} C_2[\theta, L_3] + \operatorname{Re} C_2[\theta, \theta L_2] - C_2^{(11)2}/2\pi$, the fixed-ratio trajectory gives ($\sigma = 1$):

x_o	$C_2^{(11)}$	$\frac{1}{4}C_2^{(22)}$	$\frac{1}{3}\operatorname{Re} C_2[\theta, L_3]$	$C_2^{(4)}$	$C_2^{(4)}/C_2^{(11)}$
0.15	2.209×10^{-5}	1.204×10^{-4}	8.22×10^{-5}	2.03×10^{-4}	9.17
0.3	3.414×10^{-4}	1.653×10^{-3}	9.88×10^{-4}	2.64×10^{-3}	7.74
0.6	0.00475	0.0135	−0.00084	0.0127	2.67
1	0.0258	0.0172	−0.159	−0.141	−5.5
1.5	0.0596	0.387	−0.804	−0.420	−7.0
2	0.0423	2.611	−0.581	2.010	47.6
3	0.116	6.264	−7.57	−1.57	−13.5
4	0.958	8.324	−38.7	−32.2	−33.6

The small-x_o ratios approach $\frac{29}{3} = 9.67$. Inside the horizon the 22 and 13 contributions are of the same size and partly cancel, and the net fourth-order correction to the quadrupole changes sign several times. Away from nodes the cubic term is therefore essential: the second-order part alone would give the wrong sign at $x_o = 1$–1.5 and at 3–4. Convergence of the cubic term: at $x_o = 3$ it is stable to five digits under $16 \to 24$ μ nodes and under a finer sphere ($20 \to 28$ polar nodes) and ray ($300 \to 450$) grid. At $x_o = 4$ it is stable to five digits under $16 \to 24$ μ nodes and to four digits under a finer sphere ($24 \to 32$) and ray ($400 \to 600$) grid.

Amplitude-level recombination (exploratory). To locate the cancellations, split the second-order log-map kernel into pieces P_i and measure $\kappa = \sum_i \|P_i\|^2 / \|\sum_i P_i\|^2$. The norm is the $\ell = 2$ projection integrated over μ. $\kappa \approx 1$ means the pieces barely cancel; $\kappa \gg 1$ means they cancel heavily. The pieces are:

- the local shear products, which split into the $e^{2\zeta}$ term $2(H_a' + H_b')$ and the $H_a H_b$ term;
- the A_2' term;
- the ray displacement $X \cdot \partial\sigma$ and the direction terms $\sigma(n, e)$;
- the emission endpoint and the clock.

The results:

grouping	$x_o = 1$	3	6	12
local | A_2 | ray | endpoint | clock	11.8	1.01	12.6	21.8
$\tilde{\gamma}$ (= local $+A_2$) | ray | endpoint | clock	6.3	0.91	4.9	9.9
$\tilde{\gamma}$ integrated by parts (observer, emission, bulk) | ray $\vert\cdots$	6.6	6.0	43.7	–
$A_2 + HH + e^{2\zeta}$+direction | displacement | endpoint | clock	1.78	1.24	3.8	10.3
Eulerian: above field group $-\Psi\cdot\partial\sigma$ | $(X+\Psi)\cdot\partial\sigma$ $\vert\cdots$	1.92	1.21	1.91	3.49

Four observations follow.

1. The full second-order metric $\tilde{\gamma} = A_2 + \frac{1}{2}(H_aH_b + H_bH_a) + 2(H_a + H_b)$ (so that $\sigma^{(2)} = \frac{1}{2}\partial_t\tilde{\gamma}$) roughly halves the cancellation at every x_o. Part of it is an artefact of the exponential parametrisation $\gamma = e^{2\zeta}e^H$.

2. Grouping the $e^{2\zeta}$ shear term with the direction terms removes almost all cancellation near horizon crossing ($\kappa = 1.8$ at $x_o = 1$). In orthonormal directions $e^{\zeta}e$, the static conformal factor drops out of $\sigma(e, e)$.

3. Integrating the field term by parts makes things worse inside the horizon: the observer endpoint and the bulk term cancel.

4. At late times the residual cancellation is between the field group and the photon displacement $X\cdot\partial\sigma$. The first-order kernel tends to a static profile, $Q \to 6/k_0^2$ up to oscillations, so $H_1 \to -\frac{6}{k_0^2}(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2)g$. This is a pure static relabelling of the fluid labels, $X \to X - \frac{3}{k_0^2}\nabla g$. Its products appear both in $\tilde{\gamma}$ and in the ray displacement, and they must cancel in the observable. This suggests removing the static part of H_1 by an explicit relabelling before expanding. That would be the analogue of the cancellation of bulk displacements in large-scale-structure perturbation theory. A first test supports this. Write $\Psi_a = -\frac{i}{2}Q_a(t)\,k_a e^{ik_a\cdot X}$ for the first-order fluid displacement that removes the scalar part of H_1; it tends to $-\frac{3}{k_0^2}\nabla g$ at late times. Split the photon displacement term as $X\cdot\partial\sigma = (X+\Psi)\cdot\partial\sigma - \Psi\cdot\partial\sigma$, and group $-\Psi\cdot\partial\sigma$ with the field and direction terms. This lowers κ from 3.8 to 1.9 at $x_o = 6$ and from 10.3 to 3.5 at $x_o = 12$, and leaves it unchanged near horizon crossing (last row of the table). The test keeps only the first-order scalar displacement. Including the full first-order relabelling does not change this. That relabelling is $e = (1-\partial\Psi)(e^Y - v)$ with $v = \partial_t\Psi$, and it adds the Jacobian rotation of σ and the fluid-velocity term. Kept as separate groups, the new Jacobian and direction pieces cancel each other (7.1 against 6.5 at $x_o = 12$). Combined, they return the same $\kappa = 3.5$. The residual is between the Eulerian field group and the photon's displacement relative to the fluid. A likely reason is that only space has been relabelled. In the Eulerian spatial coordinates the conformal factor becomes $\zeta_Y = \zeta(1 - Qk^2/6)$, which still oscillates with $O(1)$ amplitude ($\to -g\cos x$) at late times. The temperature-defined time slicing is also displaced, because the radiation temperature oscillates with $O(1)$ amplitude inside the horizon. The natural next step is therefore a full change to longitudinal gauge, including the time shift, in which the potentials decay as x^{-2} inside the horizon.

The observer-centred decomposition $g = g(0) + [g - g(0)]$ was also tested. Exact dilation covariance, $\Theta[\zeta + c]$ with momenta rescaled by e^{-c}, implies two identities. The purely dilation parts of $\frac{1}{4}C^{(22)}$ and $\frac{1}{3}C^{(13)}$ combine exactly into $\frac{1}{2}\sigma^2D^2C^{(11)}$, with $D = k_0\,\mathrm{d}/\mathrm{d}k_0$ at fixed times. The dilation–centred cross terms combine into a total derivative $-2D\,\mathrm{Re}\,X$, with $X = \iint PP\,A_2^cA_1^*$. This generalises the superhorizon $8 = 4 + 4$. At finite k_0, however, $\frac{1}{2}\sigma^2D^2C^{(11)}$ is not the dominant part of $C^{(4)}$: the remainder is $-8.3\,C^{(11)}$ at $x_o = 1$ and $-67\,C^{(11)}$ at $x_o = 3$. The cross term

of Prop. 13 was then evaluated (`cross_term` in `solution/numerics/explore_fourth_order.py`), which determines the centred part $C_2^{c,4} = C_2^{(4)} - \frac{1}{2}\sigma^2 D^2 C_2^{(11)} + 2D_{k_0} X_2$. In units of $C_2^{(11)}$, fixed ratio:

x_o	$C_2^{(4)}$	dilation	cross	centred
0.15	9.17	7.89	−0.25	1.53
0.3	7.74	7.54	−0.95	1.15
0.6	2.67	6.16	−3.30	−0.19
1	−5.49	2.79	−5.95	−2.33
1.5	−7.05	−4.03	−0.18	−2.83
2	47.6	−10.4	28.0	30.0
3	−13.5	53.9	−168.5	101.1

The small-x_o entries follow the proposition: the centred part tends to $\frac{5}{3}$, and the cross term vanishes as x_o^2. Inside the horizon the three pieces are each large and cancel, as at $x_o = 3$, so the observer-centred split relocates the cancellation rather than removing it. Its value is the exact superhorizon structure, not a simplification at finite k_0.

The cubic term was evaluated by contraction, as follows. Only the q-contracted third-order field is needed, $\bar{A}_3(k,t) = \int_q P(q) A_3(k, q, -q; t)$. The third-order operator depends only on the total momentum k, and after the contraction k is the only remaining direction. Hence $\bar{A}_{3,ij} = \bar{h}_3(k,t)(\hat{k}_i\hat{k}_j - \frac{1}{3}\delta_{ij})$: the loop-averaged vector and TT responses vanish. The new third-order work is therefore a scalar problem (lapse plus scalar spatial part) with q-averaged sources. These sources, and the lower-order products in L_3, still contain the second-order vectors and tensors, which are kept.

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